

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	15.1	8.1	2.6	0.8	0.2	73.2
U2 - Beta(0.5, 0.5)	15.9	8	1.6	0.5	0.2	73.8
U2 - Beta(0.7, 0.3)	15.6	8.5	2.3	0.2	0.3	73.1
U2 - Beta(1, 1)	17	7.6	1.6	0.4	0.2	73.2
U3 - Beta(0.3, 0.7)	16.1	8.7	1.6	0.2	0.1	73.3
U3 - Beta(0.5, 0.5)	17.3	7.7	1.3	0.3	0.2	73.2
U3 - Beta(0.7, 0.3)	18	7.7	1.2	0.3	0.2	72.6
U3 - Beta(1, 1)	18.8	6.8	0.9	0.2	0	73.3
Mean administrations						
U2 - Beta(0.3, 0.7)	11.1	3.6	2.2	0.9	0.4	-
U2 - Beta(0.5, 0.5)	11.1	3.6	2.1	0.9	0.3	-
U2 - Beta(0.7, 0.3)	11	3.7	2.1	0.9	0.4	-
U2 - Beta(1, 1)	11.1	3.6	2.1	0.9	0.4	-
U3 - Beta(0.3, 0.7)	11	3.6	2.2	0.9	0.4	-
U3 - Beta(0.5, 0.5)	11.1	3.6	2.2	0.9	0.3	-
U3 - Beta(0.7, 0.3)	11	3.7	2.1	0.9	0.3	-
U3 - Beta(1, 1)	11.1	3.6	2.1	0.9	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.2	1.8	0.8	0.4	0.1	-
U2 - Beta(0.5, 0.5)	3.2	1.8	0.7	0.4	0.1	-
U2 - Beta(0.7, 0.3)	3.2	1.8	0.7	0.4	0.1	-
U2 - Beta(1, 1)	3.2	1.8	0.7	0.4	0.1	-
U3 - Beta(0.3, 0.7)	3.2	1.8	0.8	0.4	0.1	-
U3 - Beta(0.5, 0.5)	3.2	1.8	0.7	0.4	0.1	-
U3 - Beta(0.7, 0.3)	3.2	1.8	0.8	0.4	0.1	-
U3 - Beta(1, 1)	3.2	1.8	0.7	0.4	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						644.8
U2 - Beta(0.5, 0.5)						642.5
U2 - Beta(0.7, 0.3)						644.2
U2 - Beta(1, 1)						644.2
U3 - Beta(0.3, 0.7)						643.7
U3 - Beta(0.5, 0.5)						645
U3 - Beta(0.7, 0.3)						644.6
U3 - Beta(1, 1)						644
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	30.4	14.4	33.7	11.7	4.7	5.1
U2 - Beta(0.5, 0.5)	31.7	15	33.2	11.2	3.7	5.2
U2 - Beta(0.7, 0.3)	32.1	14.2	34.5	11.1	3.1	5
U2 - Beta(1, 1)	32	15.5	32.3	11.3	3.7	5.2
U3 - Beta(0.3, 0.7)	34.4	14.6	32.2	10.5	3.2	5.1
U3 - Beta(0.5, 0.5)	34.3	16.3	31.2	10.2	3	5
U3 - Beta(0.7, 0.3)	35.7	16.1	31.4	9.5	2.2	5.1
U3 - Beta(1, 1)	36	16.6	29.7	10.1	2.6	5
Mean administrations						
U2 - Beta(0.3, 0.7)	10.8	5	7.3	3.9	2.2	-
U2 - Beta(0.5, 0.5)	10.8	5.1	7.2	3.8	2.2	-
U2 - Beta(0.7, 0.3)	10.9	5.1	7.3	3.9	2.1	-
U2 - Beta(1, 1)	11	5	7.2	3.9	2.1	-
U3 - Beta(0.3, 0.7)	10.8	5	7.2	4	2.2	-
U3 - Beta(0.5, 0.5)	10.8	5.1	7.1	3.9	2.1	-
U3 - Beta(0.7, 0.3)	10.9	5.1	7.2	3.8	2.1	-
U3 - Beta(1, 1)	10.9	5.1	7.1	3.9	2.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.1	2.4	2	1	0.6	-
U2 - Beta(0.5, 0.5)	3.1	2.4	1.9	1	0.5	-
U2 - Beta(0.7, 0.3)	3.1	2.4	1.9	1	0.5	-
U2 - Beta(1, 1)	3.1	2.4	1.9	1.1	0.5	-
U3 - Beta(0.3, 0.7)	3	2.4	1.9	1.1	0.5	-
U3 - Beta(0.5, 0.5)	3.1	2.4	1.9	1.1	0.5	-
U3 - Beta(0.7, 0.3)	3.1	2.5	1.9	1	0.5	-
U3 - Beta(1, 1)	3.1	2.4	1.9	1	0.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1107.5
U2 - Beta(0.5, 0.5)						1107.9
U2 - Beta(0.7, 0.3)						1107.8
U2 - Beta(1, 1)						1106.5
U3 - Beta(0.3, 0.7)						1108.2
U3 - Beta(0.5, 0.5)						1107.9
U3 - Beta(0.7, 0.3)						1107.2
U3 - Beta(1, 1)						1107.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.9	38.9	21.9	7.4	2.9	18
U2 - Beta(0.5, 0.5)	10.5	39.5	21	8.2	2.7	18.1
U2 - Beta(0.7, 0.3)	10.8	39.7	21.1	7.5	2.7	18.2
U2 - Beta(1, 1)	11.2	38.6	22.1	7.7	2.3	18.1
U3 - Beta(0.3, 0.7)	11.7	41	19.7	7.3	2.1	18.2
U3 - Beta(0.5, 0.5)	12.6	42.7	18.9	5.7	1.9	18.2
U3 - Beta(0.7, 0.3)	12.2	43.7	18.5	5.6	1.5	18.5
U3 - Beta(1, 1)	14.4	42.7	17.1	6.4	1.4	18
Mean administrations						
U2 - Beta(0.3, 0.7)	7.8	7.4	6.2	3.6	2.1	-
U2 - Beta(0.5, 0.5)	7.8	7.5	6.1	3.6	2	-
U2 - Beta(0.7, 0.3)	7.9	7.6	6.1	3.5	2	-
U2 - Beta(1, 1)	7.8	7.5	6.1	3.6	2	-
U3 - Beta(0.3, 0.7)	7.8	7.2	6.2	3.7	2.1	-
U3 - Beta(0.5, 0.5)	7.8	7.6	6	3.6	2	-
U3 - Beta(0.7, 0.3)	7.9	7.5	6.1	3.5	2	-
U3 - Beta(1, 1)	7.8	7.5	6.2	3.6	2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2	3.2	1.7	1.2	0.6	-
U2 - Beta(0.5, 0.5)	2	3.3	1.7	1.1	0.6	-
U2 - Beta(0.7, 0.3)	2	3.2	1.7	1.1	0.6	-
U2 - Beta(1, 1)	2	3.3	1.7	1.1	0.6	-
U3 - Beta(0.3, 0.7)	2	3.2	1.7	1.2	0.6	-
U3 - Beta(0.5, 0.5)	2	3.3	1.7	1.1	0.6	-
U3 - Beta(0.7, 0.3)	2.1	3.2	1.7	1.1	0.6	-
U3 - Beta(1, 1)	2	3.3	1.7	1.1	0.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1012.2
U2 - Beta(0.5, 0.5)						1011.7
U2 - Beta(0.7, 0.3)						1011.2
U2 - Beta(1, 1)						1011.9
U3 - Beta(0.3, 0.7)						1011.4
U3 - Beta(0.5, 0.5)						1011.7
U3 - Beta(0.7, 0.3)						1011.1
U3 - Beta(1, 1)						1012.3

Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6.1	6.2	21.7	43.2	20.1	2.7
U2 - Beta(0.5, 0.5)	6.4	7.2	21.6	42.5	19.4	2.9
U2 - Beta(0.7, 0.3)	6.6	7.7	18.8	44.2	20	2.7
U2 - Beta(1, 1)	7.3	7.7	18.9	43.9	19.2	3
U3 - Beta(0.3, 0.7)	8.2	8.4	20.3	42.3	17.9	2.9
U3 - Beta(0.5, 0.5)	10.6	8.8	18.7	41.8	17.2	2.9
U3 - Beta(0.7, 0.3)	11.2	9.2	17.9	42.1	16.6	3
U3 - Beta(1, 1)	12.8	9.1	17.7	42	15.6	2.8
Mean administrations						
U2 - Beta(0.3, 0.7)	4.3	4.1	6.5	8	6.5	-
U2 - Beta(0.5, 0.5)	4.3	4.2	6.5	8	6.5	-
U2 - Beta(0.7, 0.3)	4.3	4.2	6.5	8	6.4	-
U2 - Beta(1, 1)	4.3	4.2	6.4	8.1	6.4	-
U3 - Beta(0.3, 0.7)	4.3	4.1	6.5	7.9	6.6	-
U3 - Beta(0.5, 0.5)	4.3	4.1	6.4	8	6.5	-
U3 - Beta(0.7, 0.3)	4.3	4.2	6.3	8.1	6.4	-
U3 - Beta(1, 1)	4.4	4.2	6.3	8.1	6.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.6	2.7	2	2.7	1.5	-
U2 - Beta(0.5, 0.5)	0.6	2.7	2	2.6	1.5	-
U2 - Beta(0.7, 0.3)	0.6	2.8	2	2.6	1.5	-
U2 - Beta(1, 1)	0.6	2.8	2	2.6	1.5	-
U3 - Beta(0.3, 0.7)	0.6	2.7	2	2.6	1.5	-
U3 - Beta(0.5, 0.5)	0.6	2.7	2	2.6	1.5	-
U3 - Beta(0.7, 0.3)	0.6	2.7	2	2.6	1.5	-
U3 - Beta(1, 1)	0.6	2.7	1.9	2.6	1.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1098.4
U2 - Beta(0.5, 0.5)						1097.8
U2 - Beta(0.7, 0.3)						1096.5
U2 - Beta(1, 1)						1097.7
U3 - Beta(0.3, 0.7)						1097.5
U3 - Beta(0.5, 0.5)						1097.3
U3 - Beta(0.7, 0.3)						1097.3
U3 - Beta(1, 1)						1096.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.9	35.4	24.7	18.5	12	5.5
U2 - Beta(0.5, 0.5)	3.7	36.3	24.8	18.5	11.2	5.5
U2 - Beta(0.7, 0.3)	4.7	37.6	25.3	18.1	8.8	5.5
U2 - Beta(1, 1)	5.1	37	25.8	17.5	9.1	5.5
U3 - Beta(0.3, 0.7)	6.1	41.4	23.6	16.2	7.2	5.5
U3 - Beta(0.5, 0.5)	6.9	42.5	22.5	16.3	6.3	5.5
U3 - Beta(0.7, 0.3)	8.9	43.1	21.1	15.9	5.5	5.5
U3 - Beta(1, 1)	11	42.9	20.6	14.3	5.7	5.5
Mean administrations						
U2 - Beta(0.3, 0.7)	4.5	6.3	6.6	5.8	5.8	-
U2 - Beta(0.5, 0.5)	4.5	6.5	6.5	5.8	5.6	-
U2 - Beta(0.7, 0.3)	4.5	6.5	6.5	5.8	5.6	-
U2 - Beta(1, 1)	4.5	6.5	6.6	5.7	5.6	-
U3 - Beta(0.3, 0.7)	4.5	6.4	6.6	5.7	5.8	-
U3 - Beta(0.5, 0.5)	4.5	6.5	6.6	5.7	5.6	-
U3 - Beta(0.7, 0.3)	4.5	6.6	6.6	5.8	5.5	-
U3 - Beta(1, 1)	4.6	6.5	6.5	5.7	5.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.4	2.1	2.2	2	-
U2 - Beta(0.5, 0.5)	0.7	3.5	2.1	2.2	1.9	-
U2 - Beta(0.7, 0.3)	0.7	3.5	2.1	2.2	1.9	-
U2 - Beta(1, 1)	0.7	3.4	2.1	2.2	1.9	-
U3 - Beta(0.3, 0.7)	0.7	3.4	2.1	2.2	2	-
U3 - Beta(0.5, 0.5)	0.7	3.4	2.1	2.2	1.9	-
U3 - Beta(0.7, 0.3)	0.7	3.5	2.1	2.2	1.9	-
U3 - Beta(1, 1)	0.7	3.5	2.1	2.2	1.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1021.9
U2 - Beta(0.5, 0.5)						1021.1
U2 - Beta(0.7, 0.3)						1020.6
U2 - Beta(1, 1)						1021.2
U3 - Beta(0.3, 0.7)						1022
U3 - Beta(0.5, 0.5)						1020.3
U3 - Beta(0.7, 0.3)						1020.7
U3 - Beta(1, 1)						1020.3
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	66.3	12.9	3.8	1.5	0.5	15
U2 - Beta(0.5, 0.5)	66.7	13	3.6	1.3	0.4	15
U2 - Beta(0.7, 0.3)	65.7	13.6	4.5	1	0.5	14.7
U2 - Beta(1, 1)	66	13.5	4.1	1.1	0.5	14.8
U3 - Beta(0.3, 0.7)	68.4	12	3.3	1.1	0.4	14.8
U3 - Beta(0.5, 0.5)	67.3	13.2	3.3	1	0.4	14.8
U3 - Beta(0.7, 0.3)	68.2	12.6	3.1	0.8	0.4	14.9
U3 - Beta(1, 1)	69	11.7	3.2	0.8	0.4	14.9
Mean administrations						
U2 - Beta(0.3, 0.7)	19.2	3.9	2.5	1.1	0.4	-
U2 - Beta(0.5, 0.5)	19.2	4	2.5	1.1	0.4	-
U2 - Beta(0.7, 0.3)	19.2	4	2.5	1.1	0.4	-
U2 - Beta(1, 1)	19.2	4	2.5	1.1	0.4	-
U3 - Beta(0.3, 0.7)	19.2	3.9	2.6	1.1	0.4	-
U3 - Beta(0.5, 0.5)	19.1	4	2.6	1.1	0.4	-
U3 - Beta(0.7, 0.3)	19.2	3.9	2.5	1.2	0.4	-
U3 - Beta(1, 1)	19.2	3.9	2.6	1.1	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.3	1.5	0.7	0.4	0.1	-
U2 - Beta(0.5, 0.5)	5.2	1.5	0.7	0.4	0.1	-
U2 - Beta(0.7, 0.3)	5.3	1.6	0.7	0.4	0.1	-
U2 - Beta(1, 1)	5.2	1.6	0.7	0.3	0.1	-
U3 - Beta(0.3, 0.7)	5.3	1.5	0.8	0.4	0.1	-
U3 - Beta(0.5, 0.5)	5.2	1.6	0.8	0.4	0.1	-
U3 - Beta(0.7, 0.3)	5.3	1.5	0.7	0.4	0.1	-
U3 - Beta(1, 1)	5.2	1.5	0.8	0.4	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1057.4
U2 - Beta(0.5, 0.5)						1056.7
U2 - Beta(0.7, 0.3)						1057.6
U2 - Beta(1, 1)						1057.2
U3 - Beta(0.3, 0.7)						1058.4
U3 - Beta(0.5, 0.5)						1057.1
U3 - Beta(0.7, 0.3)						1057.1
U3 - Beta(1, 1)						1056.5

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	22.7	35.7	19.2	11.4	5.1	5.9
U2 - Beta(0.5, 0.5)	22.1	37.1	20.3	11	3.6	5.9
U2 - Beta(0.7, 0.3)	22	37.7	20.8	9.8	3.8	5.9
U2 - Beta(1, 1)	23.2	37	19	10.3	4.6	5.9
U3 - Beta(0.3, 0.7)	24	39.1	18.6	9.1	3.3	5.9
U3 - Beta(0.5, 0.5)	25.1	39.8	18.1	8.5	2.7	5.8
U3 - Beta(0.7, 0.3)	24.7	40.4	17.4	9.4	2.3	5.8
U3 - Beta(1, 1)	26.2	41.2	16.6	7.6	2.6	5.8
Mean administrations						
U2 - Beta(0.3, 0.7)	10.1	6.8	5.7	3.8	2.6	-
U2 - Beta(0.5, 0.5)	10.1	7	5.5	3.8	2.5	-
U2 - Beta(0.7, 0.3)	10.2	7.1	5.6	3.8	2.4	-
U2 - Beta(1, 1)	10.1	7.1	5.6	3.7	2.6	-
U3 - Beta(0.3, 0.7)	10.1	6.8	5.7	3.8	2.6	-
U3 - Beta(0.5, 0.5)	10.1	7	5.6	3.8	2.5	-
U3 - Beta(0.7, 0.3)	10.2	7	5.6	3.7	2.5	-
U3 - Beta(1, 1)	10.1	7	5.6	3.7	2.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.8	2.8	1.4	1.1	0.7	-
U2 - Beta(0.5, 0.5)	2.8	2.8	1.4	1.1	0.7	-
U2 - Beta(0.7, 0.3)	2.8	2.8	1.4	1.1	0.7	-
U2 - Beta(1, 1)	2.8	2.8	1.4	1	0.7	-
U3 - Beta(0.3, 0.7)	2.8	2.8	1.4	1.1	0.7	-
U3 - Beta(0.5, 0.5)	2.8	2.8	1.4	1.1	0.7	-
U3 - Beta(0.7, 0.3)	2.8	2.8	1.4	1.1	0.7	-
U3 - Beta(1, 1)	2.8	2.8	1.4	1.1	0.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1118.4
U2 - Beta(0.5, 0.5)						1118.8
U2 - Beta(0.7, 0.3)						1118.6
U2 - Beta(1, 1)						1118.6
U3 - Beta(0.3, 0.7)						1119
U3 - Beta(0.5, 0.5)						1119.8
U3 - Beta(0.7, 0.3)						1118.6
U3 - Beta(1, 1)						1119.1
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	25.6	9.1	3.5	1.2	0.3	60.3
U2 - Beta(0.5, 0.5)	26.2	8.5	3.2	1	0.4	60.7
U2 - Beta(0.7, 0.3)	26.6	9.3	2.5	1.2	0.3	60.1
U2 - Beta(1, 1)	26.6	9.2	2.2	1.3	0.2	60.5
U3 - Beta(0.3, 0.7)	26.5	8.9	2.7	1.1	0.4	60.4
U3 - Beta(0.5, 0.5)	27.4	8.8	2.3	0.9	0.3	60.3
U3 - Beta(0.7, 0.3)	28.3	7.9	2	1	0.3	60.5
U3 - Beta(1, 1)	28.8	8.3	1.1	1	0.3	60.5
Mean administrations						
U2 - Beta(0.3, 0.7)	13.4	3.6	1.9	0.8	0.3	-
U2 - Beta(0.5, 0.5)	13.4	3.6	1.9	0.8	0.3	-
U2 - Beta(0.7, 0.3)	13.4	3.6	1.9	0.8	0.3	-
U2 - Beta(1, 1)	13.4	3.6	1.9	0.8	0.3	-
U3 - Beta(0.3, 0.7)	13.4	3.6	1.9	0.8	0.3	-
U3 - Beta(0.5, 0.5)	13.4	3.6	1.9	0.8	0.3	-
U3 - Beta(0.7, 0.3)	13.4	3.6	1.8	0.8	0.3	-
U3 - Beta(1, 1)	13.4	3.6	1.8	0.8	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.9	1.6	0.6	0.3	0.1	-
U2 - Beta(0.5, 0.5)	3.9	1.6	0.6	0.3	0.1	-
U2 - Beta(0.7, 0.3)	3.9	1.7	0.6	0.3	0.1	-
U2 - Beta(1, 1)	3.9	1.7	0.6	0.3	0.1	-
U3 - Beta(0.3, 0.7)	3.9	1.6	0.6	0.3	0.1	-
U3 - Beta(0.5, 0.5)	3.9	1.7	0.6	0.3	0.1	-
U3 - Beta(0.7, 0.3)	3.9	1.7	0.6	0.3	0.1	-
U3 - Beta(1, 1)	3.9	1.7	0.6	0.3	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						735.3
U2 - Beta(0.5, 0.5)						733.4
U2 - Beta(0.7, 0.3)						734.1
U2 - Beta(1, 1)						733.4
U3 - Beta(0.3, 0.7)						733.4
U3 - Beta(0.5, 0.5)						732.9
U3 - Beta(0.7, 0.3)						733.3
U3 - Beta(1, 1)						732.7

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.4	8.9	16.3	14.4	27	26
U2 - Beta(0.5, 0.5)	7.7	8.6	17.8	13.6	26.4	25.9
U2 - Beta(0.7, 0.3)	9.5	9.6	16.6	13.2	25.3	25.8
U2 - Beta(1, 1)	10.4	9.3	15.9	12.2	26	26.2
U3 - Beta(0.3, 0.7)	12.3	9.5	16.2	11.2	24.6	26.2
U3 - Beta(0.5, 0.5)	13.9	9.8	15.1	12.7	22.5	26
U3 - Beta(0.7, 0.3)	17.2	9.9	14.1	11.8	20.8	26.2
U3 - Beta(1, 1)	18.9	11	13.4	10	20.3	26.4
Mean administrations						
U2 - Beta(0.3, 0.7)	6.9	4	5.3	4	5.5	-
U2 - Beta(0.5, 0.5)	6.9	4.1	5.3	4.1	5.2	-
U2 - Beta(0.7, 0.3)	6.9	4.2	5.3	4	5.2	-
U2 - Beta(1, 1)	6.9	4.2	5.2	4	5.2	-
U3 - Beta(0.3, 0.7)	6.8	4	5.4	4	5.4	-
U3 - Beta(0.5, 0.5)	6.8	4.1	5.3	4.1	5.3	-
U3 - Beta(0.7, 0.3)	6.9	4.2	5.2	4	5.2	-
U3 - Beta(1, 1)	6.9	4.2	5.2	4	5.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.7	2.4	1.8	1.6	1.6	-
U2 - Beta(0.5, 0.5)	1.7	2.4	1.8	1.6	1.5	-
U2 - Beta(0.7, 0.3)	1.7	2.4	1.8	1.6	1.5	-
U2 - Beta(1, 1)	1.7	2.5	1.8	1.6	1.5	-
U3 - Beta(0.3, 0.7)	1.7	2.4	1.9	1.6	1.6	-
U3 - Beta(0.5, 0.5)	1.7	2.4	1.8	1.6	1.5	-
U3 - Beta(0.7, 0.3)	1.7	2.4	1.8	1.6	1.5	-
U3 - Beta(1, 1)	1.7	2.5	1.8	1.6	1.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						909.6
U2 - Beta(0.5, 0.5)						908.6
U2 - Beta(0.7, 0.3)						907.8
U2 - Beta(1, 1)						906.6
U3 - Beta(0.3, 0.7)						909
U3 - Beta(0.5, 0.5)						908.7
U3 - Beta(0.7, 0.3)						907.8
U3 - Beta(1, 1)						906.4

Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6.3	28.2	12.5	4.4	2.4	46.2
U2 - Beta(0.5, 0.5)	6.4	29.1	12	4.3	1.9	46.3
U2 - Beta(0.7, 0.3)	6.4	28.6	12	4.5	2	46.5
U2 - Beta(1, 1)	6.2	29.5	11.4	4.3	2.1	46.5
U3 - Beta(0.3, 0.7)	7	29.4	10.9	4.6	1.8	46.3
U3 - Beta(0.5, 0.5)	6.9	30.5	10.7	4	1.6	46.3
U3 - Beta(0.7, 0.3)	6.7	32.1	9.2	4.1	1.4	46.5
U3 - Beta(1, 1)	6.4	31.3	11.4	3.4	1.2	46.3
Mean administrations						
U2 - Beta(0.3, 0.7)	8.4	5.8	4.2	2.2	1.2	-
U2 - Beta(0.5, 0.5)	8.4	5.9	4.1	2.2	1.1	-
U2 - Beta(0.7, 0.3)	8.4	5.9	4	2.2	1.1	-
U2 - Beta(1, 1)	8.4	5.9	4	2.2	1.2	-
U3 - Beta(0.3, 0.7)	8.4	5.7	4.1	2.2	1.2	-
U3 - Beta(0.5, 0.5)	8.4	6	4.1	2.2	1.2	-
U3 - Beta(0.7, 0.3)	8.4	5.9	4	2.2	1.1	-
U3 - Beta(1, 1)	8.4	5.9	4	2.1	1.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.2	2.5	1.1	0.7	0.3	-
U2 - Beta(0.5, 0.5)	2.2	2.5	1.1	0.7	0.3	-
U2 - Beta(0.7, 0.3)	2.2	2.5	1.1	0.6	0.3	-
U2 - Beta(1, 1)	2.2	2.5	1.1	0.7	0.3	-
U3 - Beta(0.3, 0.7)	2.2	2.5	1.1	0.7	0.3	-
U3 - Beta(0.5, 0.5)	2.2	2.5	1.1	0.6	0.3	-
U3 - Beta(0.7, 0.3)	2.2	2.5	1.1	0.7	0.3	-
U3 - Beta(1, 1)	2.2	2.5	1.1	0.6	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						815.4
U2 - Beta(0.5, 0.5)						817.3
U2 - Beta(0.7, 0.3)						815.6
U2 - Beta(1, 1)						815.6
U3 - Beta(0.3, 0.7)						816.3
U3 - Beta(0.5, 0.5)						817.3
U3 - Beta(0.7, 0.3)						815.4
U3 - Beta(1, 1)						816.2

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	59.3	20.4	4.5	0.6	0	15.2
U2 - Beta(0.5, 0.5)	58.1	20.8	4.6	1	0.1	15.4
U2 - Beta(0.7, 0.3)	57.9	21.1	4.7	0.9	0	15.4
U2 - Beta(1, 1)	57.5	21.6	4.3	1	0	15.6
U3 - Beta(0.3, 0.7)	61.4	19.8	2.9	0.5	0	15.4
U3 - Beta(0.5, 0.5)	60.4	19.2	4.4	0.6	0	15.4
U3 - Beta(0.7, 0.3)	61.4	19.7	3.1	0.6	0	15.2
U3 - Beta(1, 1)	61.9	19	2.8	0.9	0	15.4
Mean administrations						
U2 - Beta(0.3, 0.7)	17	5.8	3.5	1	0.2	-
U2 - Beta(0.5, 0.5)	16.8	5.9	3.5	1.1	0.2	-
U2 - Beta(0.7, 0.3)	16.9	5.8	3.5	1.1	0.2	-
U2 - Beta(1, 1)	16.9	5.9	3.4	1.1	0.2	-
U3 - Beta(0.3, 0.7)	16.9	5.8	3.5	1.1	0.2	-
U3 - Beta(0.5, 0.5)	16.9	5.7	3.6	1	0.2	-
U3 - Beta(0.7, 0.3)	17	5.8	3.4	1.1	0.2	-
U3 - Beta(1, 1)	17	5.8	3.4	1.1	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.2	2.5	1.1	0.4	0	-
U2 - Beta(0.5, 0.5)	5.1	2.5	1.1	0.4	0.1	-
U2 - Beta(0.7, 0.3)	5.2	2.5	1.1	0.4	0	-
U2 - Beta(1, 1)	5.2	2.6	1.1	0.4	0.1	-
U3 - Beta(0.3, 0.7)	5.2	2.5	1.1	0.4	0	-
U3 - Beta(0.5, 0.5)	5.2	2.5	1.1	0.4	0.1	-
U3 - Beta(0.7, 0.3)	5.2	2.5	1.1	0.4	0.1	-
U3 - Beta(1, 1)	5.2	2.5	1.1	0.4	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1003.6
U2 - Beta(0.5, 0.5)						1003.3
U2 - Beta(0.7, 0.3)						1002.6
U2 - Beta(1, 1)						1002.8
U3 - Beta(0.3, 0.7)						1001.9
U3 - Beta(0.5, 0.5)						1002.3
U3 - Beta(0.7, 0.3)						1002.4
U3 - Beta(1, 1)						1002.2
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.7	12.1	10.2	37.4	24.4	5.2
U2 - Beta(0.5, 0.5)	12.2	11.4	9.6	37.6	23.8	5.4
U2 - Beta(0.7, 0.3)	12.8	12	9.7	35.5	24.8	5.2
U2 - Beta(1, 1)	13.7	11.2	8.9	36.2	24.4	5.6
U3 - Beta(0.3, 0.7)	17.1	11.9	9	33.5	23.2	5.3
U3 - Beta(0.5, 0.5)	18.8	11.5	7.7	34.7	21.8	5.5
U3 - Beta(0.7, 0.3)	19	12.8	7.3	34.5	21.1	5.3
U3 - Beta(1, 1)	23.6	11.1	5.6	33.6	20.6	5.5
Mean administrations						
U2 - Beta(0.3, 0.7)	4.6	4.5	5.2	7.4	7.4	-
U2 - Beta(0.5, 0.5)	4.7	4.5	5.2	7.5	7.3	-
U2 - Beta(0.7, 0.3)	4.7	4.6	5.1	7.3	7.4	-
U2 - Beta(1, 1)	4.7	4.6	5	7.3	7.4	-
U3 - Beta(0.3, 0.7)	4.7	4.5	5.2	7.3	7.4	-
U3 - Beta(0.5, 0.5)	4.7	4.5	5.3	7.4	7.2	-
U3 - Beta(0.7, 0.3)	4.7	4.6	5.1	7.4	7.3	-
U3 - Beta(1, 1)	4.7	4.6	5.1	7.3	7.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.8	2.8	1.7	2.7	1.6	-
U2 - Beta(0.5, 0.5)	0.8	2.8	1.6	2.8	1.6	-
U2 - Beta(0.7, 0.3)	0.8	2.9	1.6	2.7	1.6	-
U2 - Beta(1, 1)	0.8	2.9	1.6	2.7	1.7	-
U3 - Beta(0.3, 0.7)	0.8	2.8	1.7	2.7	1.6	-
U3 - Beta(0.5, 0.5)	0.8	2.9	1.7	2.7	1.6	-
U3 - Beta(0.7, 0.3)	0.8	2.9	1.6	2.7	1.6	-
U3 - Beta(1, 1)	0.8	2.9	1.6	2.7	1.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1078.7
U2 - Beta(0.5, 0.5)						1074.5
U2 - Beta(0.7, 0.3)						1075.9
U2 - Beta(1, 1)						1075.2
U3 - Beta(0.3, 0.7)						1075.9
U3 - Beta(0.5, 0.5)						1074
U3 - Beta(0.7, 0.3)						1074
U3 - Beta(1, 1)						1074.3

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.9	30.2	14.1	4.6	1.9	41.3
U2 - Beta(0.5, 0.5)	8.3	29.4	13.8	5.1	2	41.4
U2 - Beta(0.7, 0.3)	8.6	30.5	13.6	4.2	2	41.1
U2 - Beta(1, 1)	8.5	29.7	14.4	4.4	1.7	41.3
U3 - Beta(0.3, 0.7)	9	32.4	11.2	4.4	1.6	41.4
U3 - Beta(0.5, 0.5)	9.5	30.2	12.9	4.6	1.3	41.5
U3 - Beta(0.7, 0.3)	9.6	31.6	12	4.1	1.4	41.3
U3 - Beta(1, 1)	11.9	31.3	11	3.6	1	41.2
Mean administrations						
U2 - Beta(0.3, 0.7)	8.2	6	4.6	2.4	1.3	-
U2 - Beta(0.5, 0.5)	8.2	6.2	4.4	2.4	1.2	-
U2 - Beta(0.7, 0.3)	8.2	6.2	4.4	2.4	1.2	-
U2 - Beta(1, 1)	8.2	6.2	4.4	2.4	1.2	-
U3 - Beta(0.3, 0.7)	8.2	6.1	4.6	2.4	1.3	-
U3 - Beta(0.5, 0.5)	8.2	6.2	4.5	2.4	1.2	-
U3 - Beta(0.7, 0.3)	8.2	6.2	4.4	2.4	1.2	-
U3 - Beta(1, 1)	8.2	6.2	4.4	2.4	1.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.1	2.7	1.4	0.8	0.4	-
U2 - Beta(0.5, 0.5)	2.2	2.8	1.3	0.8	0.4	-
U2 - Beta(0.7, 0.3)	2.1	2.8	1.3	0.8	0.4	-
U2 - Beta(1, 1)	2.2	2.8	1.3	0.8	0.4	-
U3 - Beta(0.3, 0.7)	2.1	2.8	1.4	0.8	0.4	-
U3 - Beta(0.5, 0.5)	2.1	2.8	1.4	0.8	0.4	-
U3 - Beta(0.7, 0.3)	2.2	2.8	1.3	0.8	0.4	-
U3 - Beta(1, 1)	2.2	2.8	1.3	0.8	0.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						823.4
U2 - Beta(0.5, 0.5)						822
U2 - Beta(0.7, 0.3)						823.9
U2 - Beta(1, 1)						823.3
U3 - Beta(0.3, 0.7)						823.3
U3 - Beta(0.5, 0.5)						822.3
U3 - Beta(0.7, 0.3)						823.4
U3 - Beta(1, 1)						824.6
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14.1	12.4	15.3	28.7	16.8	12.7
U2 - Beta(0.5, 0.5)	16.2	11.7	13.7	29	16.6	12.8
U2 - Beta(0.7, 0.3)	17.4	13.5	13.3	26.5	17	12.3
U2 - Beta(1, 1)	18.8	14.1	12.2	26.8	15.7	12.4
U3 - Beta(0.3, 0.7)	21.3	14.6	10.7	27.1	13.6	12.7
U3 - Beta(0.5, 0.5)	24.8	14.3	9.8	26.3	11.9	12.9
U3 - Beta(0.7, 0.3)	25.6	16.5	10.4	22.7	11.8	13
U3 - Beta(1, 1)	30.2	15.3	9.2	22.9	9.6	12.8
Mean administrations						
U2 - Beta(0.3, 0.7)	5.8	5	5.8	6.1	5.4	-
U2 - Beta(0.5, 0.5)	5.8	5.1	5.7	6.2	5.2	-
U2 - Beta(0.7, 0.3)	6	5.2	5.6	6.2	5.1	-
U2 - Beta(1, 1)	6	5.3	5.6	6.1	5.1	-
U3 - Beta(0.3, 0.7)	5.8	5	5.7	6	5.4	-
U3 - Beta(0.5, 0.5)	5.9	5.1	5.7	6.2	5.2	-
U3 - Beta(0.7, 0.3)	6	5.3	5.6	6.2	5.1	-
U3 - Beta(1, 1)	6	5.4	5.6	6	5.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.3	2.9	2	2.4	1.6	-
U2 - Beta(0.5, 0.5)	1.3	3	2	2.4	1.5	-
U2 - Beta(0.7, 0.3)	1.3	3	1.9	2.4	1.5	-
U2 - Beta(1, 1)	1.3	3.1	1.9	2.4	1.5	-
U3 - Beta(0.3, 0.7)	1.3	3	2	2.4	1.6	-
U3 - Beta(0.5, 0.5)	1.3	3	1.9	2.4	1.5	-
U3 - Beta(0.7, 0.3)	1.3	3	1.9	2.4	1.5	-
U3 - Beta(1, 1)	1.4	3.1	1.9	2.3	1.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						988
U2 - Beta(0.5, 0.5)						984.3
U2 - Beta(0.7, 0.3)						982.5
U2 - Beta(1, 1)						983.4
U3 - Beta(0.3, 0.7)						986.6
U3 - Beta(0.5, 0.5)						984.3
U3 - Beta(0.7, 0.3)						983.7
U3 - Beta(1, 1)						983.2

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	40	25.2	11.7	4.4	0.8	17.9
U2 - Beta(0.5, 0.5)	43.2	23.6	11.2	3.2	0.9	17.9
U2 - Beta(0.7, 0.3)	43.2	22.9	11.6	3.3	0.8	18.2
U2 - Beta(1, 1)	43.8	24.2	10	3.2	0.7	18.1
U3 - Beta(0.3, 0.7)	50	21.9	8.1	2.1	0.1	17.8
U3 - Beta(0.5, 0.5)	52.3	20.1	6.9	2.3	0.4	18
U3 - Beta(0.7, 0.3)	52.8	19.6	6.8	2	0.7	18.1
U3 - Beta(1, 1)	56.1	18.9	5.1	1.3	0.4	18.2
Mean administrations						
U2 - Beta(0.3, 0.7)	10	6.4	5.9	3.4	1.7	-
U2 - Beta(0.5, 0.5)	10	6.5	5.8	3.3	1.7	-
U2 - Beta(0.7, 0.3)	10	6.7	5.8	3.3	1.6	-
U2 - Beta(1, 1)	10.1	6.8	5.7	3.2	1.7	-
U3 - Beta(0.3, 0.7)	10	6.4	5.9	3.3	1.8	-
U3 - Beta(0.5, 0.5)	10	6.6	5.8	3.3	1.7	-
U3 - Beta(0.7, 0.3)	10.1	6.8	5.7	3.2	1.7	-
U3 - Beta(1, 1)	10.2	6.6	5.7	3.2	1.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3	3.3	2	1.4	0.6	-
U2 - Beta(0.5, 0.5)	3	3.4	2	1.4	0.7	-
U2 - Beta(0.7, 0.3)	3	3.4	2	1.4	0.6	-
U2 - Beta(1, 1)	3	3.5	2	1.4	0.6	-
U3 - Beta(0.3, 0.7)	3	3.3	2	1.4	0.7	-
U3 - Beta(0.5, 0.5)	3	3.4	2	1.4	0.6	-
U3 - Beta(0.7, 0.3)	3	3.4	2	1.4	0.6	-
U3 - Beta(1, 1)	3.1	3.4	2	1.4	0.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						933.5
U2 - Beta(0.5, 0.5)						932.8
U2 - Beta(0.7, 0.3)						932.9
U2 - Beta(1, 1)						933.9
U3 - Beta(0.3, 0.7)						934.4
U3 - Beta(0.5, 0.5)						934.1
U3 - Beta(0.7, 0.3)						933.6
U3 - Beta(1, 1)						931.7
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	31.8	24	15.6	16.1	12.4	0.1
U2 - Beta(0.5, 0.5)	31.3	23	16.7	16	12.9	0.1
U2 - Beta(0.7, 0.3)	32.4	23.7	17.7	14.7	11.4	0.1
U2 - Beta(1, 1)	32.8	26.5	15.8	14.7	10.1	0.1
U3 - Beta(0.3, 0.7)	38.2	26.2	14.9	13.9	6.7	0.1
U3 - Beta(0.5, 0.5)	40.1	24.6	15.8	12.3	7.1	0.1
U3 - Beta(0.7, 0.3)	41.8	26	14.1	11.5	6.5	0.1
U3 - Beta(1, 1)	44.1	28.4	12.7	9.5	5.2	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	6	5.4	5.3	5.7	7.6	-
U2 - Beta(0.5, 0.5)	6.1	5.4	5.5	5.8	7.3	-
U2 - Beta(0.7, 0.3)	6	5.5	5.5	5.9	7.1	-
U2 - Beta(1, 1)	6	5.5	5.5	5.8	7.1	-
U3 - Beta(0.3, 0.7)	6.1	5.2	5.4	5.7	7.5	-
U3 - Beta(0.5, 0.5)	6	5.5	5.5	5.8	7.3	-
U3 - Beta(0.7, 0.3)	6	5.5	5.5	5.8	7.1	-
U3 - Beta(1, 1)	6	5.5	5.4	5.8	7.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.3	3.1	1.9	2.6	2.9	-
U2 - Beta(0.5, 0.5)	1.3	3.1	1.9	2.6	2.8	-
U2 - Beta(0.7, 0.3)	1.2	3.1	1.9	2.6	2.7	-
U2 - Beta(1, 1)	1.3	3.1	1.9	2.6	2.7	-
U3 - Beta(0.3, 0.7)	1.3	3	1.9	2.6	2.9	-
U3 - Beta(0.5, 0.5)	1.3	3.1	1.9	2.6	2.8	-
U3 - Beta(0.7, 0.3)	1.3	3.1	1.9	2.6	2.7	-
U3 - Beta(1, 1)	1.2	3.1	1.9	2.6	2.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1009.8
U2 - Beta(0.5, 0.5)						1009
U2 - Beta(0.7, 0.3)						1009.1
U2 - Beta(1, 1)						1008.2
U3 - Beta(0.3, 0.7)						1009.7
U3 - Beta(0.5, 0.5)						1008.4
U3 - Beta(0.7, 0.3)						1008.7
U3 - Beta(1, 1)						1009

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	30.3	50.3	16.4	2	0	1
U2 - Beta(0.5, 0.5)	30	49.3	17.7	2.1	0	0.9
U2 - Beta(0.7, 0.3)	29.9	48.2	18.6	2.3	0	1
U2 - Beta(1, 1)	30.3	49.5	17.3	1.8	0.1	1
U3 - Beta(0.3, 0.7)	33.8	48.7	15.2	1.3	0.1	0.9
U3 - Beta(0.5, 0.5)	33.4	48.9	14.7	1.9	0.1	1
U3 - Beta(0.7, 0.3)	33.7	48.4	15	1.8	0.1	1
U3 - Beta(1, 1)	33.9	49.7	13.9	1.5	0	1
Mean administrations						
U2 - Beta(0.3, 0.7)	10	9.7	7.4	2.4	0.4	-
U2 - Beta(0.5, 0.5)	10.1	9.7	7.2	2.4	0.4	-
U2 - Beta(0.7, 0.3)	10.1	9.8	7.1	2.4	0.4	-
U2 - Beta(1, 1)	10	9.8	7.2	2.4	0.4	-
U3 - Beta(0.3, 0.7)	9.9	9.6	7.4	2.4	0.4	-
U3 - Beta(0.5, 0.5)	10.1	9.7	7.3	2.3	0.4	-
U3 - Beta(0.7, 0.3)	10.1	9.8	7.1	2.4	0.4	-
U3 - Beta(1, 1)	10	9.8	7.2	2.3	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.4	3.4	1.7	0.6	0.1	-
U2 - Beta(0.5, 0.5)	2.5	3.4	1.6	0.6	0.1	-
U2 - Beta(0.7, 0.3)	2.5	3.4	1.6	0.6	0.1	-
U2 - Beta(1, 1)	2.5	3.4	1.6	0.6	0.1	-
U3 - Beta(0.3, 0.7)	2.4	3.3	1.7	0.6	0.1	-
U3 - Beta(0.5, 0.5)	2.5	3.4	1.6	0.6	0.1	-
U3 - Beta(0.7, 0.3)	2.5	3.4	1.6	0.6	0.1	-
U3 - Beta(1, 1)	2.5	3.4	1.6	0.6	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1192.6
U2 - Beta(0.5, 0.5)						1191.4
U2 - Beta(0.7, 0.3)						1190.4
U2 - Beta(1, 1)						1191.5
U3 - Beta(0.3, 0.7)						1192.2
U3 - Beta(0.5, 0.5)						1190.8
U3 - Beta(0.7, 0.3)						1190.9
U3 - Beta(1, 1)						1192.5
Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.6	7.8	9.5	14.8	58.3	1
U2 - Beta(0.5, 0.5)	9.1	8.3	9.6	13.6	58.3	1.1
U2 - Beta(0.7, 0.3)	11	8.6	9.3	12.3	57.8	1
U2 - Beta(1, 1)	12.1	7.6	9.3	11.3	58.8	0.9
U3 - Beta(0.3, 0.7)	12.8	11	10	11.7	53.6	0.9
U3 - Beta(0.5, 0.5)	15.4	11.5	10.5	11.2	50.4	1
U3 - Beta(0.7, 0.3)	18	12.6	9.5	11.7	47.3	0.9
U3 - Beta(1, 1)	20.4	13	8.3	11	46.3	1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.6	3.7	4.5	5.5	12.4	-
U2 - Beta(0.5, 0.5)	3.7	3.9	4.6	5.5	12.2	-
U2 - Beta(0.7, 0.3)	3.7	4	4.6	5.6	12	-
U2 - Beta(1, 1)	3.7	3.9	4.6	5.5	12.2	-
U3 - Beta(0.3, 0.7)	3.6	3.7	4.5	5.6	12.4	-
U3 - Beta(0.5, 0.5)	3.7	3.8	4.6	5.6	12.1	-
U3 - Beta(0.7, 0.3)	3.7	3.9	4.6	5.5	12	-
U3 - Beta(1, 1)	3.8	3.9	4.6	5.5	12.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.3	2.6	1.5	2.5	3.4	-
U2 - Beta(0.5, 0.5)	0.3	2.7	1.5	2.5	3.4	-
U2 - Beta(0.7, 0.3)	0.3	2.7	1.5	2.5	3.3	-
U2 - Beta(1, 1)	0.3	2.7	1.5	2.5	3.4	-
U3 - Beta(0.3, 0.7)	0.3	2.6	1.5	2.5	3.4	-
U3 - Beta(0.5, 0.5)	0.3	2.7	1.5	2.5	3.4	-
U3 - Beta(0.7, 0.3)	0.3	2.7	1.5	2.5	3.4	-
U3 - Beta(1, 1)	0.3	2.7	1.5	2.5	3.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1075.3
U2 - Beta(0.5, 0.5)						1073.3
U2 - Beta(0.7, 0.3)						1071.1
U2 - Beta(1, 1)						1072.1
U3 - Beta(0.3, 0.7)						1077.3
U3 - Beta(0.5, 0.5)						1071.8
U3 - Beta(0.7, 0.3)						1071.6
U3 - Beta(1, 1)						1071.6

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14.1	51.8	18.1	1.6	0.1	14.3
U2 - Beta(0.5, 0.5)	16.1	50.6	17	1.7	0	14.6
U2 - Beta(0.7, 0.3)	14.6	52	16.9	2.2	0	14.3
U2 - Beta(1, 1)	15.7	51.3	16.1	2	0.1	14.8
U3 - Beta(0.3, 0.7)	17.8	51.3	14.6	1.6	0.1	14.6
U3 - Beta(0.5, 0.5)	18.8	51	14.1	1.6	0	14.5
U3 - Beta(0.7, 0.3)	20.1	49.4	14.2	1.8	0.1	14.4
U3 - Beta(1, 1)	20.3	50.2	13.5	1.7	0	14.3
Mean administrations						
U2 - Beta(0.3, 0.7)	7.5	10.1	7.2	2.2	0.4	-
U2 - Beta(0.5, 0.5)	7.6	10.2	7	2.2	0.4	-
U2 - Beta(0.7, 0.3)	7.6	10.3	6.9	2.2	0.4	-
U2 - Beta(1, 1)	7.6	10.2	6.9	2.2	0.4	-
U3 - Beta(0.3, 0.7)	7.5	10.1	7.1	2.3	0.4	-
U3 - Beta(0.5, 0.5)	7.6	10.2	6.9	2.2	0.4	-
U3 - Beta(0.7, 0.3)	7.6	10.3	6.9	2.2	0.4	-
U3 - Beta(1, 1)	7.7	10.3	6.9	2.2	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.8	4.4	2	0.7	0.1	-
U2 - Beta(0.5, 0.5)	1.9	4.4	2	0.8	0.1	-
U2 - Beta(0.7, 0.3)	1.9	4.4	1.9	0.8	0.1	-
U2 - Beta(1, 1)	1.9	4.4	2	0.8	0.1	-
U3 - Beta(0.3, 0.7)	1.8	4.4	2	0.8	0.1	-
U3 - Beta(0.5, 0.5)	1.9	4.4	1.9	0.8	0.1	-
U3 - Beta(0.7, 0.3)	1.9	4.5	1.9	0.8	0.1	-
U3 - Beta(1, 1)	1.9	4.4	1.9	0.8	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1007.6
U2 - Beta(0.5, 0.5)						1006.2
U2 - Beta(0.7, 0.3)						1004.9
U2 - Beta(1, 1)						1004.7
U3 - Beta(0.3, 0.7)						1007.7
U3 - Beta(0.5, 0.5)						1005.2
U3 - Beta(0.7, 0.3)						1004.8
U3 - Beta(1, 1)						1006
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.6	42.1	30	9.9	1.4	6
U2 - Beta(0.5, 0.5)	10.5	44	28.8	9.1	1.4	6.2
U2 - Beta(0.7, 0.3)	11.7	44.2	26.3	10.3	1.4	6.1
U2 - Beta(1, 1)	13.3	43.4	27	9	1.2	6.1
U3 - Beta(0.3, 0.7)	15.4	46.8	24	6.8	0.8	6.2
U3 - Beta(0.5, 0.5)	16.4	46.5	22.6	7.6	0.8	6.1
U3 - Beta(0.7, 0.3)	17.7	45.7	22.2	7.8	0.6	6
U3 - Beta(1, 1)	21.5	46.2	19.4	6.4	0.5	6
Mean administrations						
U2 - Beta(0.3, 0.7)	5.1	8.5	8.4	4.6	2.3	-
U2 - Beta(0.5, 0.5)	5.1	8.6	8.2	4.7	2.3	-
U2 - Beta(0.7, 0.3)	5.2	8.8	8.1	4.5	2.3	-
U2 - Beta(1, 1)	5.2	8.7	8.1	4.5	2.2	-
U3 - Beta(0.3, 0.7)	5	8.4	8.4	4.7	2.3	-
U3 - Beta(0.5, 0.5)	5.1	8.6	8.3	4.6	2.2	-
U3 - Beta(0.7, 0.3)	5.2	8.8	8.1	4.5	2.2	-
U3 - Beta(1, 1)	5.2	8.8	8.1	4.5	2.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.9	4.2	2.6	1.7	0.8	-
U2 - Beta(0.5, 0.5)	1	4.2	2.5	1.7	0.7	-
U2 - Beta(0.7, 0.3)	1	4.3	2.5	1.7	0.7	-
U2 - Beta(1, 1)	1	4.3	2.5	1.7	0.7	-
U3 - Beta(0.3, 0.7)	0.9	4.2	2.6	1.7	0.7	-
U3 - Beta(0.5, 0.5)	1	4.3	2.5	1.7	0.7	-
U3 - Beta(0.7, 0.3)	1	4.3	2.5	1.7	0.7	-
U3 - Beta(1, 1)	1	4.3	2.5	1.7	0.7	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1027.1
U2 - Beta(0.5, 0.5)						1025.2
U2 - Beta(0.7, 0.3)						1025.2
U2 - Beta(1, 1)						1026.4
U3 - Beta(0.3, 0.7)						1028.7
U3 - Beta(0.5, 0.5)						1024.5
U3 - Beta(0.7, 0.3)						1026.1
U3 - Beta(1, 1)						1024.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	62	1.3	0	0	0	36.7
U2 - Beta(0.5, 0.5)	61.8	1.6	0	0	0	36.6
U2 - Beta(0.7, 0.3)	61.9	1.8	0	0	0	36.3
U2 - Beta(1, 1)	61.8	1.8	0	0	0	36.4
U3 - Beta(0.3, 0.7)	62.3	1.3	0	0	0	36.4
U3 - Beta(0.5, 0.5)	62.1	1.3	0	0	0	36.6
U3 - Beta(0.7, 0.3)	61.9	1.3	0	0	0	36.8
U3 - Beta(1, 1)	62.2	1.2	0	0	0	36.6
Mean administrations						
U2 - Beta(0.3, 0.7)	21.1	2.1	0.6	0	0	-
U2 - Beta(0.5, 0.5)	21.2	2.1	0.6	0	0	-
U2 - Beta(0.7, 0.3)	21.1	2.1	0.6	0	0	-
U2 - Beta(1, 1)	21.1	2.1	0.6	0	0	-
U3 - Beta(0.3, 0.7)	21.1	2.1	0.6	0	0	-
U3 - Beta(0.5, 0.5)	21.1	2.1	0.6	0	0	-
U3 - Beta(0.7, 0.3)	21.1	2.1	0.6	0	0	-
U3 - Beta(1, 1)	21.1	2.1	0.6	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.3	0.6	0.1	0	0	-
U2 - Beta(0.5, 0.5)	4.3	0.6	0.1	0	0	-
U2 - Beta(0.7, 0.3)	4.3	0.6	0.1	0	0	-
U2 - Beta(1, 1)	4.3	0.6	0.1	0	0	-
U3 - Beta(0.3, 0.7)	4.3	0.6	0.1	0	0	-
U3 - Beta(0.5, 0.5)	4.3	0.6	0.1	0	0	-
U3 - Beta(0.7, 0.3)	4.3	0.6	0.1	0	0	-
U3 - Beta(1, 1)	4.3	0.6	0.1	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1038.1
U2 - Beta(0.5, 0.5)						1040.8
U2 - Beta(0.7, 0.3)						1041.6
U2 - Beta(1, 1)						1040
U3 - Beta(0.3, 0.7)						1040.1
U3 - Beta(0.5, 0.5)						1040.2
U3 - Beta(0.7, 0.3)						1038.4
U3 - Beta(1, 1)						1039.6

Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	22.8	53.4	11.7	2.2	0	9.9
U2 - Beta(0.5, 0.5)	22.7	52.9	13	1.8	0	9.6
U2 - Beta(0.7, 0.3)	22.8	52.5	13.1	2.1	0	9.5
U2 - Beta(1, 1)	22.7	52.7	12.9	1.9	0	9.8
U3 - Beta(0.3, 0.7)	25.6	52.3	10.6	1.6	0	9.9
U3 - Beta(0.5, 0.5)	25.8	52.2	10.8	1.4	0	9.8
U3 - Beta(0.7, 0.3)	28.3	50.3	10.2	1.9	0	9.3
U3 - Beta(1, 1)	31.3	48.2	9.8	1.6	0	9.1
Mean administrations						
U2 - Beta(0.3, 0.7)	7.5	12.1	6.5	1.8	0.3	-
U2 - Beta(0.5, 0.5)	7.6	12.1	6.3	1.8	0.3	-
U2 - Beta(0.7, 0.3)	7.7	12.2	6.3	1.7	0.3	-
U2 - Beta(1, 1)	7.8	12.1	6.3	1.7	0.3	-
U3 - Beta(0.3, 0.7)	7.5	12	6.6	1.8	0.3	-
U3 - Beta(0.5, 0.5)	7.6	12.1	6.5	1.7	0.3	-
U3 - Beta(0.7, 0.3)	7.7	12.1	6.4	1.7	0.3	-
U3 - Beta(1, 1)	7.7	12.1	6.4	1.7	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.8	5.2	1.9	0.6	0.1	-
U2 - Beta(0.5, 0.5)	1.8	5.3	1.8	0.6	0.1	-
U2 - Beta(0.7, 0.3)	1.8	5.3	1.8	0.6	0.1	-
U2 - Beta(1, 1)	1.9	5.2	1.8	0.5	0.1	-
U3 - Beta(0.3, 0.7)	1.8	5.2	1.9	0.6	0.1	-
U3 - Beta(0.5, 0.5)	1.8	5.2	1.9	0.6	0.1	-
U3 - Beta(0.7, 0.3)	1.9	5.3	1.9	0.6	0.1	-
U3 - Beta(1, 1)	1.9	5.2	1.8	0.6	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1023.7
U2 - Beta(0.5, 0.5)						1021.9
U2 - Beta(0.7, 0.3)						1022.1
U2 - Beta(1, 1)						1022.5
U3 - Beta(0.3, 0.7)						1024.8
U3 - Beta(0.5, 0.5)						1021.8
U3 - Beta(0.7, 0.3)						1021.8
U3 - Beta(1, 1)						1023.7

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.4	3.7	0.4	0	0	88.5
U2 - Beta(0.5, 0.5)	7.2	3.9	0.4	0	0	88.5
U2 - Beta(0.7, 0.3)	7.2	4	0.4	0	0	88.4
U2 - Beta(1, 1)	7.3	4.1	0.3	0	0	88.3
U3 - Beta(0.3, 0.7)	7.3	3.8	0.3	0	0	88.6
U3 - Beta(0.5, 0.5)	7.3	3.7	0.4	0	0	88.6
U3 - Beta(0.7, 0.3)	7.4	3.9	0.3	0	0	88.4
U3 - Beta(1, 1)	7.5	3.9	0.3	0	0	88.3
Mean administrations						
U2 - Beta(0.3, 0.7)	11.7	2.5	0.6	0	0	-
U2 - Beta(0.5, 0.5)	11.8	2.5	0.6	0	0	-
U2 - Beta(0.7, 0.3)	11.8	2.5	0.6	0	0	-
U2 - Beta(1, 1)	11.8	2.5	0.6	0	0	-
U3 - Beta(0.3, 0.7)	11.7	2.5	0.6	0	0	-
U3 - Beta(0.5, 0.5)	11.8	2.5	0.6	0	0	-
U3 - Beta(0.7, 0.3)	11.8	2.5	0.6	0	0	-
U3 - Beta(1, 1)	11.7	2.5	0.6	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.3	1.3	0.2	0	0	-
U2 - Beta(0.5, 0.5)	3.3	1.3	0.2	0	0	-
U2 - Beta(0.7, 0.3)	3.3	1.3	0.2	0	0	-
U2 - Beta(1, 1)	3.3	1.3	0.2	0	0	-
U3 - Beta(0.3, 0.7)	3.3	1.3	0.2	0	0	-
U3 - Beta(0.5, 0.5)	3.3	1.3	0.2	0	0	-
U3 - Beta(0.7, 0.3)	3.3	1.3	0.2	0	0	-
U3 - Beta(1, 1)	3.3	1.3	0.2	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						547.8
U2 - Beta(0.5, 0.5)						547.3
U2 - Beta(0.7, 0.3)						547.3
U2 - Beta(1, 1)						547.2
U3 - Beta(0.3, 0.7)						547.4
U3 - Beta(0.5, 0.5)						547.8
U3 - Beta(0.7, 0.3)						547.4
U3 - Beta(1, 1)						547.4
Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	1.1	1.3	3.9	45	48.5	0.2
U2 - Beta(0.5, 0.5)	1.2	1.1	5.8	45.4	46.4	0.1
U2 - Beta(0.7, 0.3)	1.6	1.5	6.1	46.7	44	0.1
U2 - Beta(1, 1)	1.4	1.3	5.7	45.6	45.9	0.1
U3 - Beta(0.3, 0.7)	2.4	2.6	6.1	46.3	42.5	0.1
U3 - Beta(0.5, 0.5)	2.8	2.2	7.5	47.3	40.1	0.1
U3 - Beta(0.7, 0.3)	3.2	3.4	7.6	48.3	37.4	0.1
U3 - Beta(1, 1)	3.3	3.5	7.2	49.7	36.2	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.2	3.2	3.8	8.1	11.7	-
U2 - Beta(0.5, 0.5)	3.3	3.2	3.8	8.4	11.3	-
U2 - Beta(0.7, 0.3)	3.3	3.3	3.8	8.5	11.1	-
U2 - Beta(1, 1)	3.3	3.3	3.8	8.4	11.3	-
U3 - Beta(0.3, 0.7)	3.2	3.2	3.8	8.1	11.7	-
U3 - Beta(0.5, 0.5)	3.3	3.2	3.8	8.4	11.3	-
U3 - Beta(0.7, 0.3)	3.3	3.3	3.8	8.4	11.2	-
U3 - Beta(1, 1)	3.3	3.3	3.8	8.4	11.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.1	2.4	1.2	3.2	3.2	-
U2 - Beta(0.5, 0.5)	0.1	2.5	1.2	3.3	3.1	-
U2 - Beta(0.7, 0.3)	0.1	2.5	1.2	3.3	3.1	-
U2 - Beta(1, 1)	0.1	2.5	1.2	3.3	3.1	-
U3 - Beta(0.3, 0.7)	0.1	2.4	1.2	3.2	3.2	-
U3 - Beta(0.5, 0.5)	0.1	2.5	1.2	3.3	3.1	-
U3 - Beta(0.7, 0.3)	0.1	2.5	1.2	3.3	3.1	-
U3 - Beta(1, 1)	0.1	2.5	1.2	3.3	3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1098.6
U2 - Beta(0.5, 0.5)						1096.3
U2 - Beta(0.7, 0.3)						1095
U2 - Beta(1, 1)						1096.7
U3 - Beta(0.3, 0.7)						1098.5
U3 - Beta(0.5, 0.5)						1096
U3 - Beta(0.7, 0.3)						1095.3
U3 - Beta(1, 1)						1096.3

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	32.5	27.8	32.5	4.8	0	2.4
U2 - Beta(0.5, 0.5)	31.9	26.8	33.5	5.3	0	2.5
U2 - Beta(0.7, 0.3)	30.6	28.6	32.6	5.7	0	2.5
U2 - Beta(1, 1)	32.4	28.2	31.6	5.3	0	2.5
U3 - Beta(0.3, 0.7)	36.8	28	28.6	4	0	2.6
U3 - Beta(0.5, 0.5)	37.2	27.6	28.7	4.2	0	2.3
U3 - Beta(0.7, 0.3)	36.7	28.1	28.3	4.3	0	2.6
U3 - Beta(1, 1)	38.6	28.3	25.9	4.6	0	2.6
Mean administrations						
U2 - Beta(0.3, 0.7)	9.1	7.8	9.4	2.9	0.5	-
U2 - Beta(0.5, 0.5)	9.1	8.1	9.2	2.8	0.5	-
U2 - Beta(0.7, 0.3)	9.2	7.9	9.2	2.8	0.5	-
U2 - Beta(1, 1)	9.2	7.9	9.2	2.8	0.5	-
U3 - Beta(0.3, 0.7)	9	7.7	9.5	2.9	0.5	-
U3 - Beta(0.5, 0.5)	9.1	8	9.2	2.7	0.5	-
U3 - Beta(0.7, 0.3)	9.2	7.9	9.2	2.8	0.5	-
U3 - Beta(1, 1)	9.3	7.9	9.1	2.9	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.4	3.5	2.5	0.9	0.1	-
U2 - Beta(0.5, 0.5)	2.4	3.6	2.5	0.8	0.1	-
U2 - Beta(0.7, 0.3)	2.5	3.5	2.5	0.9	0.1	-
U2 - Beta(1, 1)	2.5	3.5	2.5	0.9	0.1	-
U3 - Beta(0.3, 0.7)	2.4	3.4	2.6	0.9	0.1	-
U3 - Beta(0.5, 0.5)	2.4	3.6	2.5	0.9	0.1	-
U3 - Beta(0.7, 0.3)	2.5	3.5	2.5	0.9	0.1	-
U3 - Beta(1, 1)	2.5	3.5	2.5	0.9	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1115.4
U2 - Beta(0.5, 0.5)						1113.3
U2 - Beta(0.7, 0.3)						1114.8
U2 - Beta(1, 1)						1113.9
U3 - Beta(0.3, 0.7)						1116.9
U3 - Beta(0.5, 0.5)						1114.3
U3 - Beta(0.7, 0.3)						1115.1
U3 - Beta(1, 1)						1114.5
Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	20.9	22.9	30.5	5.6	0	20.1
U2 - Beta(0.5, 0.5)	19.2	24	30.4	6	0	20.4
U2 - Beta(0.7, 0.3)	19.5	26	27.1	7.4	0	20
U2 - Beta(1, 1)	20.7	25.2	27.4	6.3	0.1	20.3
U3 - Beta(0.3, 0.7)	23.6	24.2	26.8	5	0	20.4
U3 - Beta(0.5, 0.5)	23.1	26.4	24.4	6	0.1	20
U3 - Beta(0.7, 0.3)	25.4	26.3	21.5	6.5	0	20.3
U3 - Beta(1, 1)	26	25.5	22	6.1	0	20.4
Mean administrations						
U2 - Beta(0.3, 0.7)	8.9	6.3	8.1	3	0.5	-
U2 - Beta(0.5, 0.5)	8.9	6.5	8	2.8	0.5	-
U2 - Beta(0.7, 0.3)	9	6.6	7.9	2.8	0.5	-
U2 - Beta(1, 1)	9	6.5	7.9	2.9	0.5	-
U3 - Beta(0.3, 0.7)	8.9	6.3	8.1	3	0.5	-
U3 - Beta(0.5, 0.5)	9	6.6	7.9	2.9	0.5	-
U3 - Beta(0.7, 0.3)	8.9	6.5	7.9	3	0.5	-
U3 - Beta(1, 1)	9	6.6	7.9	2.9	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.5	3.1	2.3	1	0.1	-
U2 - Beta(0.5, 0.5)	2.5	3.2	2.3	0.9	0.1	-
U2 - Beta(0.7, 0.3)	2.5	3.2	2.3	0.9	0.1	-
U2 - Beta(1, 1)	2.5	3.2	2.2	0.9	0.1	-
U3 - Beta(0.3, 0.7)	2.5	3.1	2.3	0.9	0.1	-
U3 - Beta(0.5, 0.5)	2.5	3.2	2.3	0.9	0.1	-
U3 - Beta(0.7, 0.3)	2.5	3.2	2.3	0.9	0.1	-
U3 - Beta(1, 1)	2.5	3.2	2.3	0.9	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						986
U2 - Beta(0.5, 0.5)						984.5
U2 - Beta(0.7, 0.3)						984.6
U2 - Beta(1, 1)						985
U3 - Beta(0.3, 0.7)						985.6
U3 - Beta(0.5, 0.5)						985.3
U3 - Beta(0.7, 0.3)						984.4
U3 - Beta(1, 1)						983

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	15.2	20.7	43.1	8.9	0.7	11.4
U2 - Beta(0.5, 0.5)	15.9	20.3	40.9	10.7	0.7	11.5
U2 - Beta(0.7, 0.3)	15.1	23	39.7	10.1	0.7	11.4
U2 - Beta(1, 1)	13.8	22.4	42	9.7	0.6	11.5
U3 - Beta(0.3, 0.7)	18.9	22.1	38.6	8.4	0.4	11.6
U3 - Beta(0.5, 0.5)	20.2	22.2	36.7	8.9	0.7	11.3
U3 - Beta(0.7, 0.3)	18.6	24.3	36.9	8	0.9	11.3
U3 - Beta(1, 1)	19	24.1	36.5	8.5	0.4	11.5
Mean administrations						
U2 - Beta(0.3, 0.7)	6.9	6.2	9.5	4.1	1.4	-
U2 - Beta(0.5, 0.5)	7.1	6.3	9.4	4.1	1.4	-
U2 - Beta(0.7, 0.3)	7	6.4	9.2	4.1	1.4	-
U2 - Beta(1, 1)	7.1	6.3	9.4	4	1.4	-
U3 - Beta(0.3, 0.7)	6.9	6.2	9.5	4	1.5	-
U3 - Beta(0.5, 0.5)	7	6.3	9.5	4	1.4	-
U3 - Beta(0.7, 0.3)	7.1	6.4	9.3	4	1.4	-
U3 - Beta(1, 1)	7	6.4	9.3	4.1	1.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.7	3.1	2.7	1.2	0.3	-
U2 - Beta(0.5, 0.5)	1.7	3.2	2.7	1.2	0.3	-
U2 - Beta(0.7, 0.3)	1.7	3.2	2.6	1.2	0.3	-
U2 - Beta(1, 1)	1.8	3.1	2.7	1.2	0.4	-
U3 - Beta(0.3, 0.7)	1.7	3.1	2.7	1.2	0.4	-
U3 - Beta(0.5, 0.5)	1.8	3.1	2.7	1.2	0.3	-
U3 - Beta(0.7, 0.3)	1.7	3.2	2.7	1.2	0.4	-
U3 - Beta(1, 1)	1.7	3.2	2.6	1.2	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1052.7
U2 - Beta(0.5, 0.5)						1050.3
U2 - Beta(0.7, 0.3)						1050.2
U2 - Beta(1, 1)						1050.8
U3 - Beta(0.3, 0.7)						1053.5
U3 - Beta(0.5, 0.5)						1050.7
U3 - Beta(0.7, 0.3)						1050.6
U3 - Beta(1, 1)						1051
Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.8	35.3	7.1	0.3	0	43.5
U2 - Beta(0.5, 0.5)	14.8	35.1	6.6	0.2	0	43.3
U2 - Beta(0.7, 0.3)	14.3	35.2	6.2	0.4	0	43.9
U2 - Beta(1, 1)	14.8	35	6.1	0.4	0	43.7
U3 - Beta(0.3, 0.7)	15.3	34.8	6	0.2	0	43.7
U3 - Beta(0.5, 0.5)	15.5	35.5	5.5	0.3	0	43.2
U3 - Beta(0.7, 0.3)	15.9	35.4	5.1	0.2	0	43.4
U3 - Beta(1, 1)	15.8	35.5	4.8	0.3	0	43.6
Mean administrations						
U2 - Beta(0.3, 0.7)	8.7	8.3	4.3	0.9	0.1	-
U2 - Beta(0.5, 0.5)	8.7	8.4	4.3	0.9	0.1	-
U2 - Beta(0.7, 0.3)	8.9	8.3	4.2	0.9	0.1	-
U2 - Beta(1, 1)	8.9	8.3	4.2	0.9	0.1	-
U3 - Beta(0.3, 0.7)	8.7	8.3	4.3	0.9	0.1	-
U3 - Beta(0.5, 0.5)	8.8	8.4	4.2	0.9	0.1	-
U3 - Beta(0.7, 0.3)	8.8	8.4	4.1	1	0.1	-
U3 - Beta(1, 1)	8.8	8.3	4.2	0.9	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.3	3.4	1.2	0.3	0	-
U2 - Beta(0.5, 0.5)	2.3	3.4	1.2	0.3	0	-
U2 - Beta(0.7, 0.3)	2.4	3.4	1.2	0.3	0	-
U2 - Beta(1, 1)	2.4	3.4	1.2	0.3	0	-
U3 - Beta(0.3, 0.7)	2.3	3.4	1.2	0.3	0	-
U3 - Beta(0.5, 0.5)	2.3	3.4	1.2	0.3	0	-
U3 - Beta(0.7, 0.3)	2.3	3.4	1.2	0.3	0	-
U3 - Beta(1, 1)	2.4	3.4	1.2	0.3	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						830.8
U2 - Beta(0.5, 0.5)						830.3
U2 - Beta(0.7, 0.3)						829.1
U2 - Beta(1, 1)						830.5
U3 - Beta(0.3, 0.7)						829.6
U3 - Beta(0.5, 0.5)						829.2
U3 - Beta(0.7, 0.3)						830.5
U3 - Beta(1, 1)						829.7

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	68	25.3	3.7	0.2	0	2.8
U2 - Beta(0.5, 0.5)	67.2	25.6	4.2	0.3	0	2.7
U2 - Beta(0.7, 0.3)	68.5	24.3	4.2	0.3	0	2.7
U2 - Beta(1, 1)	67.3	25.5	4.1	0.4	0	2.7
U3 - Beta(0.3, 0.7)	70.2	23.7	3.2	0.2	0	2.7
U3 - Beta(0.5, 0.5)	69.8	23.7	3.4	0.3	0	2.8
U3 - Beta(0.7, 0.3)	70	23.6	3.5	0.2	0	2.7
U3 - Beta(1, 1)	71.3	22.5	3.2	0.2	0	2.8
Mean administrations						
U2 - Beta(0.3, 0.7)	17.9	7.4	3.6	0.6	0	-
U2 - Beta(0.5, 0.5)	18	7.3	3.6	0.5	0	-
U2 - Beta(0.7, 0.3)	18	7.3	3.5	0.5	0.1	-
U2 - Beta(1, 1)	18	7.3	3.6	0.6	0	-
U3 - Beta(0.3, 0.7)	18	7.3	3.6	0.6	0	-
U3 - Beta(0.5, 0.5)	18	7.3	3.6	0.5	0.1	-
U3 - Beta(0.7, 0.3)	18	7.3	3.6	0.5	0	-
U3 - Beta(1, 1)	18	7.2	3.6	0.6	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.6	2.4	1	0.2	0	-
U2 - Beta(0.5, 0.5)	4.6	2.4	1	0.2	0	-
U2 - Beta(0.7, 0.3)	4.6	2.4	1	0.2	0	-
U2 - Beta(1, 1)	4.7	2.4	0.9	0.2	0	-
U3 - Beta(0.3, 0.7)	4.7	2.4	0.9	0.2	0	-
U3 - Beta(0.5, 0.5)	4.6	2.4	1	0.2	0	-
U3 - Beta(0.7, 0.3)	4.7	2.4	1	0.2	0	-
U3 - Beta(1, 1)	4.7	2.3	1	0.2	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1178.1
U2 - Beta(0.5, 0.5)						1177.4
U2 - Beta(0.7, 0.3)						1177.3
U2 - Beta(1, 1)						1177.7
U3 - Beta(0.3, 0.7)						1178.7
U3 - Beta(0.5, 0.5)						1178
U3 - Beta(0.7, 0.3)						1177
U3 - Beta(1, 1)						1177.1
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	19.3	14.5	14.3	37.9	12.3	1.7
U2 - Beta(0.5, 0.5)	19.2	13.8	15.1	37.1	13.1	1.7
U2 - Beta(0.7, 0.3)	20	14.1	15.2	35.7	13.2	1.8
U2 - Beta(1, 1)	20.7	13.9	14.7	36.2	12.8	1.7
U3 - Beta(0.3, 0.7)	24.6	15.6	13.2	35.6	9.3	1.7
U3 - Beta(0.5, 0.5)	26.4	14.1	13.7	33.6	10.5	1.7
U3 - Beta(0.7, 0.3)	28.4	15.1	12.6	31.9	10.3	1.7
U3 - Beta(1, 1)	30.8	16.4	11.1	30.3	9.7	1.7
Mean administrations						
U2 - Beta(0.3, 0.7)	5	4.3	5.7	8.1	6.5	-
U2 - Beta(0.5, 0.5)	5	4.5	5.8	8	6.4	-
U2 - Beta(0.7, 0.3)	5.1	4.5	5.7	8	6.4	-
U2 - Beta(1, 1)	5	4.6	5.8	8	6.3	-
U3 - Beta(0.3, 0.7)	5	4.4	5.7	8.1	6.5	-
U3 - Beta(0.5, 0.5)	5	4.5	5.8	8	6.4	-
U3 - Beta(0.7, 0.3)	5	4.6	5.7	8	6.3	-
U3 - Beta(1, 1)	5	4.6	5.7	8.1	6.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.9	2.7	1.8	2.8	1.6	-
U2 - Beta(0.5, 0.5)	0.9	2.7	1.8	2.8	1.6	-
U2 - Beta(0.7, 0.3)	0.9	2.7	1.8	2.8	1.6	-
U2 - Beta(1, 1)	0.9	2.7	1.8	2.8	1.5	-
U3 - Beta(0.3, 0.7)	0.9	2.7	1.8	2.8	1.6	-
U3 - Beta(0.5, 0.5)	0.9	2.7	1.8	2.8	1.6	-
U3 - Beta(0.7, 0.3)	0.9	2.8	1.8	2.8	1.6	-
U3 - Beta(1, 1)	0.9	2.7	1.8	2.8	1.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1098.7
U2 - Beta(0.5, 0.5)						1097.5
U2 - Beta(0.7, 0.3)						1097
U2 - Beta(1, 1)						1097.2
U3 - Beta(0.3, 0.7)						1098.8
U3 - Beta(0.5, 0.5)						1097.5
U3 - Beta(0.7, 0.3)						1097.2
U3 - Beta(1, 1)						1097.3

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6.9	32.3	26.3	27	7.3	0.2
U2 - Beta(0.5, 0.5)	7.2	34.9	24.8	24.9	8	0.2
U2 - Beta(0.7, 0.3)	7.1	37	25.3	22.4	8	0.2
U2 - Beta(1, 1)	7.6	35.3	24.9	24.7	7.3	0.2
U3 - Beta(0.3, 0.7)	10.1	39.3	25.5	21.1	3.8	0.2
U3 - Beta(0.5, 0.5)	10.2	41.6	23.7	20.5	3.8	0.2
U3 - Beta(0.7, 0.3)	9.1	43.3	24.1	18	5.3	0.2
U3 - Beta(1, 1)	9.9	43.2	22.9	19.8	4	0.2
Mean administrations						
U2 - Beta(0.3, 0.7)	4.1	5.6	6.4	7.8	6.1	-
U2 - Beta(0.5, 0.5)	4.1	5.7	6.4	7.7	6	-
U2 - Beta(0.7, 0.3)	4.2	5.9	6.4	7.5	5.9	-
U2 - Beta(1, 1)	4.1	5.7	6.4	7.7	6.1	-
U3 - Beta(0.3, 0.7)	4.1	5.6	6.4	7.7	6.2	-
U3 - Beta(0.5, 0.5)	4.1	5.7	6.4	7.7	6	-
U3 - Beta(0.7, 0.3)	4.2	5.9	6.3	7.6	6	-
U3 - Beta(1, 1)	4.1	5.8	6.3	7.7	6.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.5	2.8	1.6	2	1.5	-
U2 - Beta(0.5, 0.5)	0.5	2.8	1.6	2	1.5	-
U2 - Beta(0.7, 0.3)	0.5	2.9	1.6	2	1.5	-
U2 - Beta(1, 1)	0.5	2.9	1.6	2	1.5	-
U3 - Beta(0.3, 0.7)	0.5	2.8	1.6	2	1.5	-
U3 - Beta(0.5, 0.5)	0.5	2.8	1.6	2	1.5	-
U3 - Beta(0.7, 0.3)	0.5	2.9	1.6	2	1.5	-
U3 - Beta(1, 1)	0.5	2.9	1.6	2	1.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1191
U2 - Beta(0.5, 0.5)						1189.8
U2 - Beta(0.7, 0.3)						1187.6
U2 - Beta(1, 1)						1188.9
U3 - Beta(0.3, 0.7)						1190.4
U3 - Beta(0.5, 0.5)						1188.8
U3 - Beta(0.7, 0.3)						1187.6
U3 - Beta(1, 1)						1189.3
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.6	12.9	33.9	27.3	14.4	0.9
U2 - Beta(0.5, 0.5)	8.7	13.4	34.7	27.2	15.1	0.9
U2 - Beta(0.7, 0.3)	9.8	14.5	34.1	26.7	14.1	0.8
U2 - Beta(1, 1)	11.4	15.4	34	25.9	12.5	0.8
U3 - Beta(0.3, 0.7)	13.8	18.4	34.4	23.1	9.4	0.9
U3 - Beta(0.5, 0.5)	14.6	17.8	34.6	23.2	9	0.8
U3 - Beta(0.7, 0.3)	15.9	18.9	33.3	22.3	8.7	0.9
U3 - Beta(1, 1)	19.5	19.3	31.2	22.1	7.2	0.7
Mean administrations						
U2 - Beta(0.3, 0.7)	3.8	4.1	7.6	7.1	7.2	-
U2 - Beta(0.5, 0.5)	3.9	4.3	7.7	7.1	6.9	-
U2 - Beta(0.7, 0.3)	4	4.3	7.7	7.1	6.9	-
U2 - Beta(1, 1)	4	4.4	7.7	7	6.8	-
U3 - Beta(0.3, 0.7)	3.9	4.2	7.6	7	7.2	-
U3 - Beta(0.5, 0.5)	4	4.3	7.7	7	6.9	-
U3 - Beta(0.7, 0.3)	4	4.4	7.6	7	6.9	-
U3 - Beta(1, 1)	4	4.3	7.6	7.1	6.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	2.7	2.4	2.7	2.4	-
U2 - Beta(0.5, 0.5)	0.4	2.8	2.5	2.7	2.3	-
U2 - Beta(0.7, 0.3)	0.4	2.9	2.4	2.7	2.3	-
U2 - Beta(1, 1)	0.4	2.9	2.5	2.7	2.3	-
U3 - Beta(0.3, 0.7)	0.4	2.8	2.4	2.7	2.4	-
U3 - Beta(0.5, 0.5)	0.4	2.8	2.5	2.7	2.3	-
U3 - Beta(0.7, 0.3)	0.4	2.9	2.4	2.7	2.3	-
U3 - Beta(1, 1)	0.4	2.8	2.4	2.7	2.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1059.6
U2 - Beta(0.5, 0.5)						1054.9
U2 - Beta(0.7, 0.3)						1055.7
U2 - Beta(1, 1)						1055.3
U3 - Beta(0.3, 0.7)						1059.3
U3 - Beta(0.5, 0.5)						1055
U3 - Beta(0.7, 0.3)						1055.5
U3 - Beta(1, 1)						1055.7

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.1	5.3	12.4	39.7	34.4	0.1
U2 - Beta(0.5, 0.5)	8.6	5.1	11.6	41.1	33.3	0.3
U2 - Beta(0.7, 0.3)	8.8	4.7	13.3	40.5	32.5	0.2
U2 - Beta(1, 1)	9.5	5.7	12.7	39.6	32.3	0.2
U3 - Beta(0.3, 0.7)	10.5	6.5	13.3	41	28.5	0.2
U3 - Beta(0.5, 0.5)	12.3	6.9	13.6	40	27	0.2
U3 - Beta(0.7, 0.3)	15.3	8.1	12.8	38.5	25.2	0.1
U3 - Beta(1, 1)	16.6	8.6	12.1	37.6	24.9	0.2
Mean administrations						
U2 - Beta(0.3, 0.7)	3.5	3.6	5.2	7.9	9.9	-
U2 - Beta(0.5, 0.5)	3.5	3.6	5.2	8.1	9.6	-
U2 - Beta(0.7, 0.3)	3.5	3.7	5.1	8.2	9.5	-
U2 - Beta(1, 1)	3.5	3.7	5.1	8	9.6	-
U3 - Beta(0.3, 0.7)	3.4	3.5	5.2	7.9	9.9	-
U3 - Beta(0.5, 0.5)	3.5	3.6	5.2	8.1	9.6	-
U3 - Beta(0.7, 0.3)	3.5	3.7	5.2	8.1	9.5	-
U3 - Beta(1, 1)	3.6	3.6	5.1	7.9	9.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.2	2.6	1.6	3	2.5	-
U2 - Beta(0.5, 0.5)	0.2	2.6	1.6	3.1	2.4	-
U2 - Beta(0.7, 0.3)	0.2	2.6	1.6	3.1	2.4	-
U2 - Beta(1, 1)	0.2	2.7	1.6	3	2.4	-
U3 - Beta(0.3, 0.7)	0.2	2.6	1.6	3	2.5	-
U3 - Beta(0.5, 0.5)	0.2	2.6	1.6	3.1	2.4	-
U3 - Beta(0.7, 0.3)	0.2	2.7	1.6	3	2.4	-
U3 - Beta(1, 1)	0.3	2.6	1.6	3	2.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1109.6
U2 - Beta(0.5, 0.5)						1107.4
U2 - Beta(0.7, 0.3)						1105.9
U2 - Beta(1, 1)						1106.9
U3 - Beta(0.3, 0.7)						1108.4
U3 - Beta(0.5, 0.5)						1106.9
U3 - Beta(0.7, 0.3)						1105
U3 - Beta(1, 1)						1106.8
Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11.4	12.3	47.3	18.7	6.2	4.1
U2 - Beta(0.5, 0.5)	11.9	11.7	47	19.8	5.4	4.2
U2 - Beta(0.7, 0.3)	14	11	45.9	20.4	4.5	4.2
U2 - Beta(1, 1)	13.8	11.3	46.1	19.9	4.3	4.6
U3 - Beta(0.3, 0.7)	17.6	13.3	44.6	16.9	3.6	4
U3 - Beta(0.5, 0.5)	18.2	13.3	44.3	16.1	3.6	4.5
U3 - Beta(0.7, 0.3)	21.8	13.5	41.2	15.4	3.7	4.4
U3 - Beta(1, 1)	25.4	14.4	37.2	16.1	2.7	4.2
Mean administrations						
U2 - Beta(0.3, 0.7)	4.9	5.4	10	5.4	3.6	-
U2 - Beta(0.5, 0.5)	5	5.5	9.9	5.4	3.4	-
U2 - Beta(0.7, 0.3)	5.1	5.5	9.8	5.5	3.3	-
U2 - Beta(1, 1)	5.1	5.5	9.7	5.5	3.5	-
U3 - Beta(0.3, 0.7)	4.9	5.3	10	5.4	3.6	-
U3 - Beta(0.5, 0.5)	5	5.5	9.8	5.4	3.5	-
U3 - Beta(0.7, 0.3)	5.1	5.5	9.8	5.4	3.4	-
U3 - Beta(1, 1)	5.1	5.5	9.8	5.5	3.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.9	3.1	3.1	1.9	1.1	-
U2 - Beta(0.5, 0.5)	0.9	3.2	3.1	1.9	1.1	-
U2 - Beta(0.7, 0.3)	1	3.2	3.1	1.9	1	-
U2 - Beta(1, 1)	1	3.2	3.1	1.9	1.1	-
U3 - Beta(0.3, 0.7)	0.9	3.1	3.1	1.9	1.1	-
U3 - Beta(0.5, 0.5)	0.9	3.2	3.1	1.9	1.1	-
U3 - Beta(0.7, 0.3)	1	3.2	3.1	1.9	1.1	-
U3 - Beta(1, 1)	0.9	3.2	3.1	1.9	1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1052.7
U2 - Beta(0.5, 0.5)						1049.8
U2 - Beta(0.7, 0.3)						1049.9
U2 - Beta(1, 1)						1048.2
U3 - Beta(0.3, 0.7)						1054.3
U3 - Beta(0.5, 0.5)						1050.6
U3 - Beta(0.7, 0.3)						1047.7
U3 - Beta(1, 1)						1049.5

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	10.8	39.6	33.1	11.7	4.6	0.2
U2 - Beta(0.5, 0.5)	11.6	40.2	31.4	12.3	4.3	0.2
U2 - Beta(0.7, 0.3)	11.4	39.9	30.5	13.9	4.1	0.2
U2 - Beta(1, 1)	12.7	40.3	31	11.9	3.9	0.2
U3 - Beta(0.3, 0.7)	14.3	43.3	29.3	10	2.9	0.2
U3 - Beta(0.5, 0.5)	16.1	44	27.2	9.9	2.6	0.2
U3 - Beta(0.7, 0.3)	17.5	44.7	25.1	9.6	2.9	0.2
U3 - Beta(1, 1)	18	45.3	24.3	10	2.2	0.2
Mean administrations						
U2 - Beta(0.3, 0.7)	4.5	8.1	9	5.2	3.3	-
U2 - Beta(0.5, 0.5)	4.6	8.3	8.8	5	3.3	-
U2 - Beta(0.7, 0.3)	4.7	8.4	8.7	5	3.2	-
U2 - Beta(1, 1)	4.6	8.2	8.9	5	3.2	-
U3 - Beta(0.3, 0.7)	4.6	8	9.1	5	3.3	-
U3 - Beta(0.5, 0.5)	4.6	8.2	8.9	5	3.2	-
U3 - Beta(0.7, 0.3)	4.7	8.3	8.8	5	3.1	-
U3 - Beta(1, 1)	4.7	8.2	9	5	3.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.6	3.6	2.2	1.5	0.9	-
U2 - Beta(0.5, 0.5)	0.7	3.7	2.2	1.5	0.9	-
U2 - Beta(0.7, 0.3)	0.7	3.7	2.2	1.5	0.8	-
U2 - Beta(1, 1)	0.6	3.6	2.2	1.5	0.8	-
U3 - Beta(0.3, 0.7)	0.6	3.6	2.3	1.5	0.9	-
U3 - Beta(0.5, 0.5)	0.6	3.6	2.2	1.5	0.8	-
U3 - Beta(0.7, 0.3)	0.7	3.6	2.2	1.5	0.8	-
U3 - Beta(1, 1)	0.7	3.6	2.2	1.5	0.8	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1166.8
U2 - Beta(0.5, 0.5)						1166.3
U2 - Beta(0.7, 0.3)						1165.5
U2 - Beta(1, 1)						1166.1
U3 - Beta(0.3, 0.7)						1166.8
U3 - Beta(0.5, 0.5)						1166.4
U3 - Beta(0.7, 0.3)						1166.2
U3 - Beta(1, 1)						1165.4
Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.8	36.9	16.7	6.4	2.2	24
U2 - Beta(0.5, 0.5)	14.6	36.6	16.1	7	1.8	23.9
U2 - Beta(0.7, 0.3)	15.9	38	14.5	5.8	1.7	24.1
U2 - Beta(1, 1)	15.1	39	13.6	6.4	1.8	24.1
U3 - Beta(0.3, 0.7)	17.8	38	13.1	5.6	1.3	24.2
U3 - Beta(0.5, 0.5)	19.1	37.8	12.6	5.9	1	23.6
U3 - Beta(0.7, 0.3)	20.2	39.1	10.6	5.1	1.3	23.7
U3 - Beta(1, 1)	22.8	37.6	10	4.8	1	23.8
Mean administrations						
U2 - Beta(0.3, 0.7)	7.7	7.9	5.9	2.9	1.4	-
U2 - Beta(0.5, 0.5)	7.8	8	5.7	2.9	1.4	-
U2 - Beta(0.7, 0.3)	7.8	8.1	5.6	2.8	1.4	-
U2 - Beta(1, 1)	7.8	8.1	5.6	2.8	1.4	-
U3 - Beta(0.3, 0.7)	7.6	8	5.9	2.8	1.4	-
U3 - Beta(0.5, 0.5)	7.6	8.2	5.7	2.8	1.4	-
U3 - Beta(0.7, 0.3)	7.7	8.1	5.7	2.8	1.4	-
U3 - Beta(1, 1)	7.8	8.1	5.5	2.8	1.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.9	3.8	1.9	1.1	0.4	-
U2 - Beta(0.5, 0.5)	2	3.8	1.8	1.1	0.4	-
U2 - Beta(0.7, 0.3)	2	3.9	1.8	1	0.4	-
U2 - Beta(1, 1)	2	3.9	1.7	1.1	0.4	-
U3 - Beta(0.3, 0.7)	1.9	3.8	1.9	1	0.4	-
U3 - Beta(0.5, 0.5)	1.9	3.9	1.8	1	0.4	-
U3 - Beta(0.7, 0.3)	1.9	3.9	1.8	1.1	0.4	-
U3 - Beta(1, 1)	2	3.9	1.8	1.1	0.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						915.3
U2 - Beta(0.5, 0.5)						914.5
U2 - Beta(0.7, 0.3)						914.6
U2 - Beta(1, 1)						915.8
U3 - Beta(0.3, 0.7)						915.3
U3 - Beta(0.5, 0.5)						918.1
U3 - Beta(0.7, 0.3)						916.1
U3 - Beta(1, 1)						915.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	20.2	46.7	13.1	3.1	0.7	16.2
U2 - Beta(0.5, 0.5)	20.4	45.6	14.7	2.7	0.5	16.1
U2 - Beta(0.7, 0.3)	20.7	46.3	13	3.4	0.5	16.1
U2 - Beta(1, 1)	21.9	45.8	12.5	2.5	0.6	16.7
U3 - Beta(0.3, 0.7)	24.4	44.8	11.8	2.4	0.4	16.2
U3 - Beta(0.5, 0.5)	25.5	44.6	11.2	2	0.4	16.3
U3 - Beta(0.7, 0.3)	26.9	45.6	9.1	1.9	0.4	16.1
U3 - Beta(1, 1)	30.3	43.8	8.1	1.5	0.3	16
Mean administrations						
U2 - Beta(0.3, 0.7)	7.2	10.3	6.2	2.4	0.9	-
U2 - Beta(0.5, 0.5)	7.3	10.3	6.2	2.4	0.9	-
U2 - Beta(0.7, 0.3)	7.3	10.5	6.1	2.3	0.9	-
U2 - Beta(1, 1)	7.4	10.4	6.1	2.3	0.9	-
U3 - Beta(0.3, 0.7)	7.1	10.3	6.3	2.4	0.9	-
U3 - Beta(0.5, 0.5)	7.2	10.3	6.2	2.4	0.9	-
U3 - Beta(0.7, 0.3)	7.3	10.5	6.1	2.3	0.9	-
U3 - Beta(1, 1)	7.3	10.5	6	2.3	0.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.6	4.8	1.9	1	0.3	-
U2 - Beta(0.5, 0.5)	1.7	4.8	1.9	1	0.3	-
U2 - Beta(0.7, 0.3)	1.7	4.9	1.9	0.9	0.3	-
U2 - Beta(1, 1)	1.7	4.9	1.9	0.9	0.3	-
U3 - Beta(0.3, 0.7)	1.6	4.8	2	1	0.3	-
U3 - Beta(0.5, 0.5)	1.7	4.8	1.9	0.9	0.3	-
U3 - Beta(0.7, 0.3)	1.7	4.9	1.9	0.9	0.3	-
U3 - Beta(1, 1)	1.7	4.9	1.8	0.9	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						955.7
U2 - Beta(0.5, 0.5)						955.2
U2 - Beta(0.7, 0.3)						954
U2 - Beta(1, 1)						952.7
U3 - Beta(0.3, 0.7)						955.1
U3 - Beta(0.5, 0.5)						954.7
U3 - Beta(0.7, 0.3)						953.3
U3 - Beta(1, 1)						953.4

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	14.6	11.3	2.6	1	0.4	70.1
U2 - Beta(0.5, 0.5)	16.4	11.5	2.3	0.6	0.2	69
U2 - Beta(0.7, 0.3)	16.6	11.4	2.8	0.6	0.3	68.3
U2 - Beta(1, 1)	16.3	11	2.5	0.7	0.3	69.2
U3 - Beta(0.3, 0.7)	15.4	11.4	2	0.4	0.2	70.6
U3 - Beta(0.5, 0.5)	16.7	11.4	1.8	0.4	0.1	69.6
U3 - Beta(0.7, 0.3)	15.6	12.6	2.4	0.4	0.1	68.9
U3 - Beta(1, 1)	15.1	12.2	2.8	0.4	0.1	69.4
Mean administrations						
U2 - Beta(0.3, 0.7)	10.7	5.7	2.6	1.2	0.3	-
U2 - Beta(0.5, 0.5)	10.7	5.7	2.5	1.4	0.4	-
U2 - Beta(0.7, 0.3)	10.7	5.8	2.5	1.4	0.4	-
U2 - Beta(1, 1)	10.6	5.8	2.6	1.4	0.4	-
U3 - Beta(0.3, 0.7)	10.8	5.7	2.4	1.1	0.3	-
U3 - Beta(0.5, 0.5)	10.8	5.6	2.5	1.4	0.4	-
U3 - Beta(0.7, 0.3)	10.8	5.7	2.4	1.4	0.4	-
U3 - Beta(1, 1)	10.9	5.8	2.4	1.3	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.8	2.9	0.9	0.5	0.1	-
U2 - Beta(0.5, 0.5)	2.8	2.9	0.8	0.6	0.2	-
U2 - Beta(0.7, 0.3)	2.8	3	0.8	0.6	0.2	-
U2 - Beta(1, 1)	2.8	3	0.8	0.6	0.2	-
U3 - Beta(0.3, 0.7)	2.9	2.9	0.8	0.5	0.1	-
U3 - Beta(0.5, 0.5)	2.9	2.9	0.8	0.6	0.2	-
U3 - Beta(0.7, 0.3)	2.9	3	0.8	0.6	0.1	-
U3 - Beta(1, 1)	2.9	3	0.8	0.6	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						725
U2 - Beta(0.5, 0.5)						732.2
U2 - Beta(0.7, 0.3)						735.2
U2 - Beta(1, 1)						734.1
U3 - Beta(0.3, 0.7)						719
U3 - Beta(0.5, 0.5)						731.2
U3 - Beta(0.7, 0.3)						733.4
U3 - Beta(1, 1)						732.6
Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	23.8	25	42.4	5.8	0.4	2.6
U2 - Beta(0.5, 0.5)	23	17.3	39.6	15.4	2.1	2.6
U2 - Beta(0.7, 0.3)	24.4	19.5	35.8	14.4	3.2	2.7
U2 - Beta(1, 1)	21.3	18.6	40.7	14.5	2.4	2.5
U3 - Beta(0.3, 0.7)	31.1	30.4	31.7	3.9	0.2	2.7
U3 - Beta(0.5, 0.5)	28.1	19	35.2	13.1	1.7	2.9
U3 - Beta(0.7, 0.3)	27.6	20	34.5	13	2.5	2.4
U3 - Beta(1, 1)	27.5	19.9	35.3	12.9	1.7	2.7
Mean administrations						
U2 - Beta(0.3, 0.7)	11.2	9.2	8	1.2	0.1	-
U2 - Beta(0.5, 0.5)	9	6.9	8.9	3.9	0.9	-
U2 - Beta(0.7, 0.3)	9	6.7	7.7	4.7	1.5	-
U2 - Beta(1, 1)	8.7	7.1	9	3.9	0.9	-
U3 - Beta(0.3, 0.7)	12.6	9.7	6.4	0.8	0.1	-
U3 - Beta(0.5, 0.5)	9.3	7.1	8.8	3.6	0.7	-
U3 - Beta(0.7, 0.3)	9.3	6.9	8	4.4	1.1	-
U3 - Beta(1, 1)	9.4	7.2	8.7	3.6	0.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.2	3.9	2.4	0.4	0	-
U2 - Beta(0.5, 0.5)	2.2	3.2	2.4	1.1	0.3	-
U2 - Beta(0.7, 0.3)	2.2	3.1	2	1.3	0.4	-
U2 - Beta(1, 1)	2.1	3.2	2.5	1.1	0.3	-
U3 - Beta(0.3, 0.7)	3.8	4.1	1.9	0.3	0	-
U3 - Beta(0.5, 0.5)	2.3	3.3	2.4	1.1	0.2	-
U3 - Beta(0.7, 0.3)	2.3	3.2	2.1	1.3	0.3	-
U3 - Beta(1, 1)	2.4	3.3	2.3	1.1	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1089.2
U2 - Beta(0.5, 0.5)						1125.4
U2 - Beta(0.7, 0.3)						1131.8
U2 - Beta(1, 1)						1127.5
U3 - Beta(0.3, 0.7)						1079
U3 - Beta(0.5, 0.5)						1120.4
U3 - Beta(0.7, 0.3)						1127.3
U3 - Beta(1, 1)						1120.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	4.3	61	20.5	5.8	0.6	7.8
U2 - Beta(0.5, 0.5)	7.9	44.5	24.9	13.2	2.4	7.1
U2 - Beta(0.7, 0.3)	8.3	48.4	21.7	12.6	2	7
U2 - Beta(1, 1)	8.4	44.3	24.6	12.5	2.8	7.4
U3 - Beta(0.3, 0.7)	6.4	66.2	14.6	5	0.1	7.7
U3 - Beta(0.5, 0.5)	9.6	49.1	23.1	9.2	1.7	7.3
U3 - Beta(0.7, 0.3)	10.3	50.3	20.1	10.4	1.8	7.1
U3 - Beta(1, 1)	11.2	51.6	19.4	8.6	2	7.2
Mean administrations						
U2 - Beta(0.3, 0.7)	6.9	16.3	4.4	1.1	0.2	-
U2 - Beta(0.5, 0.5)	6.1	11.1	6.8	3.8	1.1	-
U2 - Beta(0.7, 0.3)	6	9.9	6.9	4.4	1.7	-
U2 - Beta(1, 1)	6	11.2	6.9	3.7	1.1	-
U3 - Beta(0.3, 0.7)	7.3	16.9	3.6	1	0.1	-
U3 - Beta(0.5, 0.5)	6.3	11.8	6.5	3.4	0.9	-
U3 - Beta(0.7, 0.3)	6.2	10.5	6.6	4.2	1.3	-
U3 - Beta(1, 1)	6.2	12.1	6.3	3.4	0.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.5	6.2	1.4	0.4	0.1	-
U2 - Beta(0.5, 0.5)	1.1	4.6	2	1.2	0.3	-
U2 - Beta(0.7, 0.3)	1	4.3	1.9	1.4	0.5	-
U2 - Beta(1, 1)	1	4.6	2	1.2	0.3	-
U3 - Beta(0.3, 0.7)	1.7	6.4	1.1	0.3	0	-
U3 - Beta(0.5, 0.5)	1.1	4.8	1.9	1.1	0.3	-
U3 - Beta(0.7, 0.3)	1.1	4.4	1.9	1.3	0.4	-
U3 - Beta(1, 1)	1.1	4.9	1.8	1.1	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1065.8
U2 - Beta(0.5, 0.5)						1085
U2 - Beta(0.7, 0.3)						1089.3
U2 - Beta(1, 1)						1084.8
U3 - Beta(0.3, 0.7)						1062.3
U3 - Beta(0.5, 0.5)						1081.2
U3 - Beta(0.7, 0.3)						1086.2
U3 - Beta(1, 1)						1079.6
Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.1	5.3	25.7	58.1	6.5	1.3
U2 - Beta(0.5, 0.5)	5.7	5.2	16.9	53.3	17.6	1.3
U2 - Beta(0.7, 0.3)	7.1	5.7	17.5	49	19.6	1.1
U2 - Beta(1, 1)	7	5.8	15.9	53.6	16.6	1.1
U3 - Beta(0.3, 0.7)	6.2	5.8	28.4	53.8	4.3	1.5
U3 - Beta(0.5, 0.5)	8.6	7.7	16.4	51.3	14.9	1.1
U3 - Beta(0.7, 0.3)	9.2	8.1	18.2	47.1	16.3	1.1
U3 - Beta(1, 1)	9.5	8	15.8	52.6	12.8	1.3
Mean administrations						
U2 - Beta(0.3, 0.7)	4.4	5.3	9.5	9.7	1	-
U2 - Beta(0.5, 0.5)	4.1	4.5	6.7	10.6	3.9	-
U2 - Beta(0.7, 0.3)	4.1	4.5	6.6	9.7	5	-
U2 - Beta(1, 1)	4.1	4.5	6.7	10.7	3.9	-
U3 - Beta(0.3, 0.7)	4.6	5.5	9.7	9.3	0.7	-
U3 - Beta(0.5, 0.5)	4.3	4.6	6.7	10.7	3.6	-
U3 - Beta(0.7, 0.3)	4.3	4.7	6.5	9.9	4.5	-
U3 - Beta(1, 1)	4.5	4.8	6.6	10.7	3.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.1	3.3	3.1	0.3	-
U2 - Beta(0.5, 0.5)	0.5	3	1.9	3.3	1	-
U2 - Beta(0.7, 0.3)	0.5	3	1.9	3.1	1.2	-
U2 - Beta(1, 1)	0.5	3	1.9	3.4	1	-
U3 - Beta(0.3, 0.7)	0.8	3.2	3.4	3	0.2	-
U3 - Beta(0.5, 0.5)	0.5	3	1.9	3.4	0.9	-
U3 - Beta(0.7, 0.3)	0.6	3.1	1.8	3.1	1.2	-
U3 - Beta(1, 1)	0.6	3.1	1.9	3.4	0.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1068.6
U2 - Beta(0.5, 0.5)						1107.3
U2 - Beta(0.7, 0.3)						1111.9
U2 - Beta(1, 1)						1108.6
U3 - Beta(0.3, 0.7)						1062.1
U3 - Beta(0.5, 0.5)						1104.7
U3 - Beta(0.7, 0.3)						1105.3
U3 - Beta(1, 1)						1100.6

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	1.8	39	27.6	22.4	7.6	1.6
U2 - Beta(0.5, 0.5)	5.4	37.6	25.7	19.5	10.4	1.4
U2 - Beta(0.7, 0.3)	6.9	39	24	19.6	9.1	1.4
U2 - Beta(1, 1)	6.8	38.9	24.7	18.9	9.3	1.4
U3 - Beta(0.3, 0.7)	3.4	48.7	24.5	17.2	4.6	1.6
U3 - Beta(0.5, 0.5)	9	42.6	23.2	17.5	6.2	1.5
U3 - Beta(0.7, 0.3)	8.9	43.8	23.7	16	6.2	1.4
U3 - Beta(1, 1)	10.8	46.9	19.8	15.5	5.6	1.4
Mean administrations						
U2 - Beta(0.3, 0.7)	4.2	13	6.8	4.3	1.5	-
U2 - Beta(0.5, 0.5)	3.8	7.2	7	7.1	4.6	-
U2 - Beta(0.7, 0.3)	3.8	6.7	7.1	7.4	4.8	-
U2 - Beta(1, 1)	3.8	7.2	7.1	7.1	4.5	-
U3 - Beta(0.3, 0.7)	4.4	14.1	6.5	3.6	1.1	-
U3 - Beta(0.5, 0.5)	3.9	7.9	7.2	6.8	4	-
U3 - Beta(0.7, 0.3)	3.9	7.5	7	7.1	4.2	-
U3 - Beta(1, 1)	3.9	8.2	7	6.8	3.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.5	5.8	2.5	1.6	0.5	-
U2 - Beta(0.5, 0.5)	0.3	3.8	2.3	2.7	1.6	-
U2 - Beta(0.7, 0.3)	0.3	3.7	2.2	2.8	1.6	-
U2 - Beta(1, 1)	0.3	3.8	2.3	2.7	1.6	-
U3 - Beta(0.3, 0.7)	0.6	6.2	2.4	1.4	0.4	-
U3 - Beta(0.5, 0.5)	0.3	4.1	2.3	2.6	1.4	-
U3 - Beta(0.7, 0.3)	0.3	4	2.2	2.6	1.5	-
U3 - Beta(1, 1)	0.3	4.2	2.3	2.6	1.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1035.7
U2 - Beta(0.5, 0.5)						1052.3
U2 - Beta(0.7, 0.3)						1053.8
U2 - Beta(1, 1)						1051.8
U3 - Beta(0.3, 0.7)						1032.7
U3 - Beta(0.5, 0.5)						1049.9
U3 - Beta(0.7, 0.3)						1053.2
U3 - Beta(1, 1)						1049.3
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	63.7	15.3	5	0.8	0.1	15.1
U2 - Beta(0.5, 0.5)	61.9	17.6	5.1	1.9	0.5	13
U2 - Beta(0.7, 0.3)	59.8	18.6	5	2.5	0.4	13.7
U2 - Beta(1, 1)	61.2	18.3	5.2	1.9	0.5	12.9
U3 - Beta(0.3, 0.7)	67.1	14.2	3.3	0.4	0.1	14.9
U3 - Beta(0.5, 0.5)	62.7	17.8	4.2	1.5	0.4	13.4
U3 - Beta(0.7, 0.3)	64.1	17.7	3.9	1.4	0.3	12.6
U3 - Beta(1, 1)	63.2	18.4	3.3	1.6	0.4	13.1
Mean administrations						
U2 - Beta(0.3, 0.7)	20.6	5	1.4	0.3	0	-
U2 - Beta(0.5, 0.5)	17.1	6.3	2.7	1.3	0.3	-
U2 - Beta(0.7, 0.3)	16.6	6.4	2.6	1.6	0.5	-
U2 - Beta(1, 1)	17	6.3	2.7	1.4	0.3	-
U3 - Beta(0.3, 0.7)	21.3	4.8	1	0.2	0	-
U3 - Beta(0.5, 0.5)	17.6	6.2	2.4	1.2	0.3	-
U3 - Beta(0.7, 0.3)	17.1	6.3	2.7	1.4	0.3	-
U3 - Beta(1, 1)	17.6	6.3	2.4	1.2	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	5.6	2	0.5	0.1	0	-
U2 - Beta(0.5, 0.5)	4.3	2.5	0.8	0.4	0.1	-
U2 - Beta(0.7, 0.3)	4.2	2.6	0.7	0.5	0.1	-
U2 - Beta(1, 1)	4.3	2.5	0.8	0.4	0.1	-
U3 - Beta(0.3, 0.7)	5.9	1.9	0.3	0.1	0	-
U3 - Beta(0.5, 0.5)	4.5	2.5	0.7	0.4	0.1	-
U3 - Beta(0.7, 0.3)	4.3	2.5	0.8	0.5	0.1	-
U3 - Beta(1, 1)	4.5	2.5	0.7	0.4	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1058.6
U2 - Beta(0.5, 0.5)						1081.9
U2 - Beta(0.7, 0.3)						1080.3
U2 - Beta(1, 1)						1081.8
U3 - Beta(0.3, 0.7)						1057.7
U3 - Beta(0.5, 0.5)						1078.3
U3 - Beta(0.7, 0.3)						1082.7
U3 - Beta(1, 1)						1078.9

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	12.1	70.6	12.1	2.1	0.3	2.8
U2 - Beta(0.5, 0.5)	11.8	46.8	22.7	12.5	3.5	2.7
U2 - Beta(0.7, 0.3)	11.3	45.2	23.9	13.2	3.8	2.6
U2 - Beta(1, 1)	11	47.4	24.1	11.3	3.4	2.8
U3 - Beta(0.3, 0.7)	15.7	71.7	8.1	1.6	0.2	2.7
U3 - Beta(0.5, 0.5)	13.5	50.5	20.7	10.2	2.3	2.8
U3 - Beta(0.7, 0.3)	13.5	46.5	24.2	10.8	2.3	2.7
U3 - Beta(1, 1)	13.8	53.3	19.7	8.5	1.9	2.8
Mean administrations						
U2 - Beta(0.3, 0.7)	10.3	16.4	2.4	0.4	0.1	-
U2 - Beta(0.5, 0.5)	7.4	12.5	5.8	3	0.8	-
U2 - Beta(0.7, 0.3)	7.1	9.8	6.5	4.4	1.7	-
U2 - Beta(1, 1)	7.2	12.6	6	2.9	0.8	-
U3 - Beta(0.3, 0.7)	10.9	16.4	1.8	0.3	0	-
U3 - Beta(0.5, 0.5)	7.7	13.2	5.4	2.6	0.7	-
U3 - Beta(0.7, 0.3)	7.3	10.7	6.3	4	1.2	-
U3 - Beta(1, 1)	7.7	13.4	5.4	2.5	0.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.7	5.6	0.7	0.1	0	-
U2 - Beta(0.5, 0.5)	1.4	4.6	1.6	0.9	0.2	-
U2 - Beta(0.7, 0.3)	1.4	3.8	1.6	1.3	0.5	-
U2 - Beta(1, 1)	1.3	4.6	1.6	0.9	0.2	-
U3 - Beta(0.3, 0.7)	3	5.6	0.5	0.1	0	-
U3 - Beta(0.5, 0.5)	1.5	4.8	1.4	0.8	0.2	-
U3 - Beta(0.7, 0.3)	1.4	4.1	1.7	1.1	0.4	-
U3 - Beta(1, 1)	1.5	4.9	1.4	0.7	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1127.2
U2 - Beta(0.5, 0.5)						1151.9
U2 - Beta(0.7, 0.3)						1158.1
U2 - Beta(1, 1)						1153.7
U3 - Beta(0.3, 0.7)						1124.1
U3 - Beta(0.5, 0.5)						1150.1
U3 - Beta(0.7, 0.3)						1153.6
U3 - Beta(1, 1)						1149.7
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	27	10.6	4.1	2	0.4	55.9
U2 - Beta(0.5, 0.5)	28.7	12.4	3.5	1.5	0.3	53.6
U2 - Beta(0.7, 0.3)	28.2	14.1	3.4	1	0.3	53
U2 - Beta(1, 1)	27.7	13.7	3.9	1.1	0.4	53.2
U3 - Beta(0.3, 0.7)	26.1	12.3	3	1.6	0.3	56.7
U3 - Beta(0.5, 0.5)	28.2	14	2.4	1.3	0.2	53.9
U3 - Beta(0.7, 0.3)	27.8	14.2	2.9	0.9	0.3	53.9
U3 - Beta(1, 1)	28.7	14.3	2.5	1	0.2	53.3
Mean administrations						
U2 - Beta(0.3, 0.7)	13.2	5.8	2.2	0.7	0.1	-
U2 - Beta(0.5, 0.5)	12.6	6	2.3	1.2	0.3	-
U2 - Beta(0.7, 0.3)	12.6	6	2.2	1.2	0.4	-
U2 - Beta(1, 1)	12.6	6.1	2.3	1.2	0.3	-
U3 - Beta(0.3, 0.7)	13.4	5.8	2	0.7	0.1	-
U3 - Beta(0.5, 0.5)	12.9	5.8	2.2	1.1	0.3	-
U3 - Beta(0.7, 0.3)	12.9	5.9	2.2	1.2	0.3	-
U3 - Beta(1, 1)	13	6	2.1	1.1	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	3.6	2.7	0.8	0.3	0	-
U2 - Beta(0.5, 0.5)	3.3	2.8	0.7	0.4	0.1	-
U2 - Beta(0.7, 0.3)	3.4	2.8	0.7	0.5	0.1	-
U2 - Beta(1, 1)	3.3	2.9	0.7	0.4	0.1	-
U3 - Beta(0.3, 0.7)	3.7	2.7	0.7	0.3	0	-
U3 - Beta(0.5, 0.5)	3.5	2.8	0.7	0.4	0.1	-
U3 - Beta(0.7, 0.3)	3.4	2.8	0.7	0.4	0.1	-
U3 - Beta(1, 1)	3.5	2.8	0.6	0.4	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						801.7
U2 - Beta(0.5, 0.5)						818.9
U2 - Beta(0.7, 0.3)						823.8
U2 - Beta(1, 1)						823.1
U3 - Beta(0.3, 0.7)						798.7
U3 - Beta(0.5, 0.5)						820
U3 - Beta(0.7, 0.3)						823.2
U3 - Beta(1, 1)						822.1

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.9	9.1	24.5	23.6	21.3	13.6
U2 - Beta(0.5, 0.5)	10.4	11	21.2	17	27.6	12.8
U2 - Beta(0.7, 0.3)	11.6	11	21.1	16.6	26.4	13.3
U2 - Beta(1, 1)	13.4	9.7	20.9	16.8	26.6	12.6
U3 - Beta(0.3, 0.7)	11.9	12.7	23.3	21.1	16.2	14.8
U3 - Beta(0.5, 0.5)	15.4	13.2	19.8	15.1	23.3	13.2
U3 - Beta(0.7, 0.3)	17.4	12.6	20.2	14.5	22.1	13.2
U3 - Beta(1, 1)	18.4	13.1	19.3	13.7	21.9	13.6
Mean administrations						
U2 - Beta(0.3, 0.7)	6.5	5.9	7.2	5.7	2.9	-
U2 - Beta(0.5, 0.5)	6.2	5.6	6	5.4	5.2	-
U2 - Beta(0.7, 0.3)	6.2	5.7	5.8	5.2	5.5	-
U2 - Beta(1, 1)	6.1	5.8	6	5.4	5.1	-
U3 - Beta(0.3, 0.7)	6.8	6.1	7.2	5.4	2.5	-
U3 - Beta(0.5, 0.5)	6.4	5.7	5.9	5.4	4.8	-
U3 - Beta(0.7, 0.3)	6.4	5.9	5.8	5.3	5	-
U3 - Beta(1, 1)	6.6	6	5.6	5.3	4.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.4	3.3	2.6	2.2	1	-
U2 - Beta(0.5, 0.5)	1.3	3.3	1.9	2.1	1.6	-
U2 - Beta(0.7, 0.3)	1.2	3.3	1.9	2	1.6	-
U2 - Beta(1, 1)	1.2	3.4	2	2.1	1.6	-
U3 - Beta(0.3, 0.7)	1.6	3.3	2.6	2.1	0.8	-
U3 - Beta(0.5, 0.5)	1.4	3.3	1.9	2.1	1.5	-
U3 - Beta(0.7, 0.3)	1.3	3.4	1.9	2.1	1.5	-
U3 - Beta(1, 1)	1.4	3.5	1.8	2.1	1.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						978.3
U2 - Beta(0.5, 0.5)						999.7
U2 - Beta(0.7, 0.3)						1002.3
U2 - Beta(1, 1)						999.4
U3 - Beta(0.3, 0.7)						970.1
U3 - Beta(0.5, 0.5)						997
U3 - Beta(0.7, 0.3)						996.6
U3 - Beta(1, 1)						994.7

Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	5.5	44.9	11.5	2.8	0.5	34.8
U2 - Beta(0.5, 0.5)	4.7	36.3	15.9	6.5	1.8	34.8
U2 - Beta(0.7, 0.3)	4.6	38.5	15.1	6	2.1	33.7
U2 - Beta(1, 1)	4.9	36.1	16.1	6.9	1.5	34.5
U3 - Beta(0.3, 0.7)	4.9	47.4	9.6	2.4	0.2	35.5
U3 - Beta(0.5, 0.5)	5.2	38	15.1	5.5	1.1	35.1
U3 - Beta(0.7, 0.3)	5.1	39.9	14.1	5.5	1.3	34.1
U3 - Beta(1, 1)	5.6	40.4	13.9	4.8	1	34.3
Mean administrations						
U2 - Beta(0.3, 0.7)	8	13.3	2.8	0.6	0.1	-
U2 - Beta(0.5, 0.5)	7.5	9.7	4.8	2.2	0.7	-
U2 - Beta(0.7, 0.3)	7.4	9	4.9	2.8	1	-
U2 - Beta(1, 1)	7.5	9.8	4.8	2.3	0.6	-
U3 - Beta(0.3, 0.7)	8.2	13.6	2.5	0.5	0.1	-
U3 - Beta(0.5, 0.5)	7.6	10.1	4.6	2.1	0.6	-
U3 - Beta(0.7, 0.3)	7.5	9.4	4.8	2.6	0.8	-
U3 - Beta(1, 1)	7.6	10.4	4.5	2.1	0.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.7	5.2	0.9	0.2	0	-
U2 - Beta(0.5, 0.5)	1.5	4.1	1.3	0.7	0.2	-
U2 - Beta(0.7, 0.3)	1.4	3.9	1.3	0.9	0.3	-
U2 - Beta(1, 1)	1.4	4.2	1.3	0.7	0.2	-
U3 - Beta(0.3, 0.7)	1.8	5.3	0.8	0.2	0	-
U3 - Beta(0.5, 0.5)	1.5	4.3	1.2	0.7	0.2	-
U3 - Beta(0.7, 0.3)	1.5	4	1.3	0.8	0.2	-
U3 - Beta(1, 1)	1.5	4.4	1.2	0.7	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						927.6
U2 - Beta(0.5, 0.5)						944.9
U2 - Beta(0.7, 0.3)						958.9
U2 - Beta(1, 1)						947.1
U3 - Beta(0.3, 0.7)						923.2
U3 - Beta(0.5, 0.5)						944.8
U3 - Beta(0.7, 0.3)						949.7
U3 - Beta(1, 1)						947

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	52.8	25	7.5	0.7	0.1	13.9
U2 - Beta(0.5, 0.5)	57.3	23.4	4.4	0.6	0	14.3
U2 - Beta(0.7, 0.3)	56.5	23.8	5.7	0.6	0.1	13.3
U2 - Beta(1, 1)	55.9	24	5.9	0.9	0.1	13.2
U3 - Beta(0.3, 0.7)	58.7	22.2	4.1	0.7	0	14.3
U3 - Beta(0.5, 0.5)	59.9	21.1	4.5	0.6	0	13.9
U3 - Beta(0.7, 0.3)	60.1	21.2	4.3	0.6	0.1	13.7
U3 - Beta(1, 1)	60.3	21.5	3.7	0.5	0.1	13.9
Mean administrations						
U2 - Beta(0.3, 0.7)	16.1	8	3	0.8	0.1	-
U2 - Beta(0.5, 0.5)	15.4	7.8	3.4	1.4	0.2	-
U2 - Beta(0.7, 0.3)	15.4	7.9	3.3	1.4	0.2	-
U2 - Beta(1, 1)	15.2	8	3.5	1.4	0.2	-
U3 - Beta(0.3, 0.7)	17.5	7.7	2.3	0.6	0.1	-
U3 - Beta(0.5, 0.5)	15.9	7.6	3.2	1.3	0.2	-
U3 - Beta(0.7, 0.3)	15.8	7.8	3.1	1.3	0.2	-
U3 - Beta(1, 1)	15.9	7.8	3	1.3	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.8	3.4	1	0.3	0	-
U2 - Beta(0.5, 0.5)	4.4	3.5	1	0.5	0.1	-
U2 - Beta(0.7, 0.3)	4.4	3.5	1	0.5	0.1	-
U2 - Beta(1, 1)	4.3	3.6	1	0.5	0.1	-
U3 - Beta(0.3, 0.7)	5.3	3.3	0.8	0.2	0	-
U3 - Beta(0.5, 0.5)	4.6	3.5	0.9	0.5	0.1	-
U3 - Beta(0.7, 0.3)	4.6	3.5	0.9	0.5	0.1	-
U3 - Beta(1, 1)	4.6	3.5	0.9	0.5	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1019.6
U2 - Beta(0.5, 0.5)						1031.3
U2 - Beta(0.7, 0.3)						1037.1
U2 - Beta(1, 1)						1036.1
U3 - Beta(0.3, 0.7)						1016.1
U3 - Beta(0.5, 0.5)						1032.1
U3 - Beta(0.7, 0.3)						1033.1
U3 - Beta(1, 1)						1032.5
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7	9.2	8.6	60.1	12.9	2.2
U2 - Beta(0.5, 0.5)	10.7	10.7	8.3	43.9	24.3	2.1
U2 - Beta(0.7, 0.3)	11.9	11.9	8.6	40.4	25.1	2.1
U2 - Beta(1, 1)	11.7	10.5	7.5	42.6	25.6	2.1
U3 - Beta(0.3, 0.7)	10.9	10.7	7.3	58.3	10.3	2.5
U3 - Beta(0.5, 0.5)	14.5	13.6	7.1	42.4	20.4	2
U3 - Beta(0.7, 0.3)	16.8	13.7	6.7	39	21.8	2
U3 - Beta(1, 1)	17.7	13.8	6.3	39.3	20.7	2.2
Mean administrations						
U2 - Beta(0.3, 0.7)	4.6	5.6	5.9	11.4	2.2	-
U2 - Beta(0.5, 0.5)	4.4	5	5.4	9.5	5.5	-
U2 - Beta(0.7, 0.3)	4.5	5	5.5	8.6	6.3	-
U2 - Beta(1, 1)	4.4	5	5.4	9.4	5.5	-
U3 - Beta(0.3, 0.7)	4.9	5.9	5.9	11.1	1.9	-
U3 - Beta(0.5, 0.5)	4.6	5.1	5.4	9.6	5.2	-
U3 - Beta(0.7, 0.3)	4.7	5.2	5.3	8.9	5.7	-
U3 - Beta(1, 1)	4.8	5.2	5.3	9.5	5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.7	3.3	2.1	3.9	0.6	-
U2 - Beta(0.5, 0.5)	0.6	3.2	1.6	3.3	1.4	-
U2 - Beta(0.7, 0.3)	0.6	3.2	1.6	3.1	1.5	-
U2 - Beta(1, 1)	0.6	3.2	1.6	3.3	1.4	-
U3 - Beta(0.3, 0.7)	0.9	3.4	2.1	3.8	0.6	-
U3 - Beta(0.5, 0.5)	0.7	3.2	1.6	3.3	1.3	-
U3 - Beta(0.7, 0.3)	0.7	3.3	1.6	3.1	1.4	-
U3 - Beta(1, 1)	0.8	3.3	1.5	3.3	1.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1053.6
U2 - Beta(0.5, 0.5)						1084
U2 - Beta(0.7, 0.3)						1089
U2 - Beta(1, 1)						1082.8
U3 - Beta(0.3, 0.7)						1048.7
U3 - Beta(0.5, 0.5)						1079.6
U3 - Beta(0.7, 0.3)						1083.2
U3 - Beta(1, 1)						1078

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	6	37.8	16.4	5.7	1	33.1
U2 - Beta(0.5, 0.5)	6.3	36.8	17.2	6.7	1.7	31.3
U2 - Beta(0.7, 0.3)	6.8	37	16.2	6.9	1.8	31.3
U2 - Beta(1, 1)	7.4	36.6	15.6	7.1	1.6	31.7
U3 - Beta(0.3, 0.7)	6.9	40.8	13.7	4.5	0.7	33.4
U3 - Beta(0.5, 0.5)	8.5	37.6	15.9	5.1	1.3	31.6
U3 - Beta(0.7, 0.3)	7.7	38.6	15.1	5.9	1	31.7
U3 - Beta(1, 1)	9.3	38.6	14.5	4.2	1.4	32
Mean administrations						
U2 - Beta(0.3, 0.7)	8	11.6	4	1.4	0.3	-
U2 - Beta(0.5, 0.5)	7.6	9.1	5	2.9	0.9	-
U2 - Beta(0.7, 0.3)	7.5	8.9	4.9	3.1	1.1	-
U2 - Beta(1, 1)	7.5	9.1	5	3	0.9	-
U3 - Beta(0.3, 0.7)	8.2	12	3.6	1.2	0.2	-
U3 - Beta(0.5, 0.5)	7.7	9.4	4.9	2.7	0.8	-
U3 - Beta(0.7, 0.3)	7.6	9.1	4.8	3	0.9	-
U3 - Beta(1, 1)	7.7	9.6	4.7	2.6	0.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.7	4.9	1.3	0.5	0.1	-
U2 - Beta(0.5, 0.5)	1.5	4.2	1.5	1	0.3	-
U2 - Beta(0.7, 0.3)	1.5	4.1	1.4	1.1	0.3	-
U2 - Beta(1, 1)	1.5	4.2	1.5	1	0.3	-
U3 - Beta(0.3, 0.7)	1.8	5.1	1.2	0.4	0	-
U3 - Beta(0.5, 0.5)	1.6	4.3	1.4	1	0.3	-
U3 - Beta(0.7, 0.3)	1.6	4.2	1.4	1	0.3	-
U3 - Beta(1, 1)	1.6	4.4	1.4	0.9	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						912.6
U2 - Beta(0.5, 0.5)						932.7
U2 - Beta(0.7, 0.3)						938.6
U2 - Beta(1, 1)						933.8
U3 - Beta(0.3, 0.7)						907.4
U3 - Beta(0.5, 0.5)						930
U3 - Beta(0.7, 0.3)						930
U3 - Beta(1, 1)						929.9
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11.3	12.5	15	40.6	14.9	5.7
U2 - Beta(0.5, 0.5)	14.9	15.1	14.4	34.2	16	5.4
U2 - Beta(0.7, 0.3)	16.1	16.2	14.6	31.4	16.2	5.5
U2 - Beta(1, 1)	17.5	14.1	14.1	33.8	15	5.5
U3 - Beta(0.3, 0.7)	16.2	16.7	12	38.2	11	5.9
U3 - Beta(0.5, 0.5)	21.3	18.2	13.2	30.6	11.2	5.5
U3 - Beta(0.7, 0.3)	21.3	17.6	13.4	30.9	11.3	5.5
U3 - Beta(1, 1)	24.1	17	14.1	28.7	10.6	5.5
Mean administrations						
U2 - Beta(0.3, 0.7)	5.8	6.3	6.1	8.6	2.6	-
U2 - Beta(0.5, 0.5)	5.7	5.9	5.7	7.8	4.3	-
U2 - Beta(0.7, 0.3)	5.7	5.9	5.7	7.5	4.6	-
U2 - Beta(1, 1)	5.7	6	5.7	7.7	4.3	-
U3 - Beta(0.3, 0.7)	6.1	6.6	6.1	8.4	2.3	-
U3 - Beta(0.5, 0.5)	6	6.1	5.6	7.8	4	-
U3 - Beta(0.7, 0.3)	6.1	6.2	5.6	7.5	4.2	-
U3 - Beta(1, 1)	6.2	6.3	5.4	7.7	3.9	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.2	3.5	2.2	3.2	0.9	-
U2 - Beta(0.5, 0.5)	1.1	3.5	1.8	3	1.4	-
U2 - Beta(0.7, 0.3)	1.1	3.5	1.8	2.9	1.4	-
U2 - Beta(1, 1)	1.1	3.6	1.8	2.9	1.4	-
U3 - Beta(0.3, 0.7)	1.4	3.6	2.2	3.1	0.8	-
U3 - Beta(0.5, 0.5)	1.3	3.6	1.8	2.9	1.3	-
U3 - Beta(0.7, 0.3)	1.3	3.6	1.8	2.8	1.3	-
U3 - Beta(1, 1)	1.4	3.7	1.7	2.9	1.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1013
U2 - Beta(0.5, 0.5)						1028.1
U2 - Beta(0.7, 0.3)						1030.2
U2 - Beta(1, 1)						1027.7
U3 - Beta(0.3, 0.7)						1008.7
U3 - Beta(0.5, 0.5)						1024.9
U3 - Beta(0.7, 0.3)						1026.5
U3 - Beta(1, 1)						1024.6

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	38.8	28.2	13.8	6.3	1.1	11.8
U2 - Beta(0.5, 0.5)	41.9	27.2	13.1	6.5	0.4	10.9
U2 - Beta(0.7, 0.3)	41.1	25.6	13.5	7	1.2	11.6
U2 - Beta(1, 1)	41.5	24.6	14.2	7.6	1.2	10.9
U3 - Beta(0.3, 0.7)	43.3	31.3	9.3	3.6	0.4	12.1
U3 - Beta(0.5, 0.5)	46.1	28.2	10	3.3	0.5	11.9
U3 - Beta(0.7, 0.3)	47	26.5	10	3.9	0.3	12.3
U3 - Beta(1, 1)	47.1	26.3	10.7	3.4	0.3	12.2
Mean administrations						
U2 - Beta(0.3, 0.7)	9.3	8.1	6.1	4.1	1.3	-
U2 - Beta(0.5, 0.5)	9.4	8	5.7	4.1	1.6	-
U2 - Beta(0.7, 0.3)	9.4	8.1	5.8	4.1	1.6	-
U2 - Beta(1, 1)	9.4	8.2	5.9	4	1.5	-
U3 - Beta(0.3, 0.7)	10.3	8.3	5.6	3.6	1.1	-
U3 - Beta(0.5, 0.5)	9.9	8	5.5	4	1.5	-
U3 - Beta(0.7, 0.3)	10	8.1	5.5	3.9	1.4	-
U3 - Beta(1, 1)	10.1	8.2	5.4	3.9	1.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.6	4	2.2	1.6	0.5	-
U2 - Beta(0.5, 0.5)	2.6	4.2	1.9	1.7	0.5	-
U2 - Beta(0.7, 0.3)	2.7	4.2	1.9	1.7	0.5	-
U2 - Beta(1, 1)	2.6	4.2	1.9	1.7	0.5	-
U3 - Beta(0.3, 0.7)	3.1	4.1	2	1.5	0.4	-
U3 - Beta(0.5, 0.5)	2.9	4.2	1.8	1.7	0.5	-
U3 - Beta(0.7, 0.3)	2.9	4.2	1.8	1.6	0.5	-
U3 - Beta(1, 1)	2.9	4.3	1.7	1.6	0.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						979.6
U2 - Beta(0.5, 0.5)						982.4
U2 - Beta(0.7, 0.3)						984
U2 - Beta(1, 1)						983.2
U3 - Beta(0.3, 0.7)						975.4
U3 - Beta(0.5, 0.5)						981.6
U3 - Beta(0.7, 0.3)						983.2
U3 - Beta(1, 1)						982
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	20.3	21.2	19.8	22	16.7	0
U2 - Beta(0.5, 0.5)	40.6	24.4	14.6	13.1	7.3	0
U2 - Beta(0.7, 0.3)	41.2	25.6	15.3	10.5	7.4	0
U2 - Beta(1, 1)	42.9	25.3	14.7	10.9	6.2	0
U3 - Beta(0.3, 0.7)	29	24.5	19	18.8	8.7	0
U3 - Beta(0.5, 0.5)	47.1	24.2	12.6	11	5.1	0
U3 - Beta(0.7, 0.3)	46.1	27.3	13.7	8.7	4.2	0
U3 - Beta(1, 1)	50.2	25.3	13.5	7.5	3.5	0
Mean administrations						
U2 - Beta(0.3, 0.7)	7.7	6.8	5.9	6.1	3.5	-
U2 - Beta(0.5, 0.5)	4.5	4.9	6	8.3	6.3	-
U2 - Beta(0.7, 0.3)	4.4	4.8	6.2	8.1	6.4	-
U2 - Beta(1, 1)	4.6	5.1	6.2	8.1	5.9	-
U3 - Beta(0.3, 0.7)	8.3	7.2	5.9	5.6	2.9	-
U3 - Beta(0.5, 0.5)	4.7	5.1	6.2	8.2	5.7	-
U3 - Beta(0.7, 0.3)	4.7	5.3	6.4	8	5.5	-
U3 - Beta(1, 1)	4.9	5.5	6.4	8	5.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.1	3.3	2.4	2.5	1.4	-
U2 - Beta(0.5, 0.5)	0.6	2.9	2.2	3.6	2.3	-
U2 - Beta(0.7, 0.3)	0.6	2.9	2.2	3.6	2.3	-
U2 - Beta(1, 1)	0.6	3	2.3	3.5	2.2	-
U3 - Beta(0.3, 0.7)	2.4	3.5	2.4	2.4	1.2	-
U3 - Beta(0.5, 0.5)	0.7	3	2.3	3.6	2.1	-
U3 - Beta(0.7, 0.3)	0.7	3.1	2.3	3.5	2	-
U3 - Beta(1, 1)	0.8	3.2	2.3	3.5	1.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						995.4
U2 - Beta(0.5, 0.5)						1009.6
U2 - Beta(0.7, 0.3)						1011.8
U2 - Beta(1, 1)						1008.1
U3 - Beta(0.3, 0.7)						992.1
U3 - Beta(0.5, 0.5)						1008.4
U3 - Beta(0.7, 0.3)						1009
U3 - Beta(1, 1)						1005.8

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	33.6	58	7	0.6	0	0.8
U2 - Beta(0.5, 0.5)	18.3	61	17.4	2.8	0.1	0.4
U2 - Beta(0.7, 0.3)	20.6	55.9	18.3	4.4	0	0.8
U2 - Beta(1, 1)	18.8	60.7	17.2	2.7	0.1	0.5
U3 - Beta(0.3, 0.7)	38.4	56.6	4	0.3	0	0.7
U3 - Beta(0.5, 0.5)	23.2	61.2	13.2	1.9	0	0.5
U3 - Beta(0.7, 0.3)	21.6	59.5	15.7	2.5	0	0.7
U3 - Beta(1, 1)	21.5	62.5	13.5	1.9	0	0.6
Mean administrations						
U2 - Beta(0.3, 0.7)	15.5	12.7	1.6	0.1	0	-
U2 - Beta(0.5, 0.5)	9	14.1	5.4	1.3	0.2	-
U2 - Beta(0.7, 0.3)	8.9	12	6.3	2.4	0.4	-
U2 - Beta(1, 1)	8.8	14.3	5.4	1.3	0.2	-
U3 - Beta(0.3, 0.7)	16.2	12.5	1.1	0.1	0	-
U3 - Beta(0.5, 0.5)	9.5	14.5	4.8	1	0.1	-
U3 - Beta(0.7, 0.3)	9	12.9	5.8	2	0.2	-
U3 - Beta(1, 1)	9.5	14.7	4.6	1	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.6	4.2	0.4	0.1	0	-
U2 - Beta(0.5, 0.5)	1.9	4.7	1.4	0.4	0	-
U2 - Beta(0.7, 0.3)	1.9	4.2	1.4	0.7	0.1	-
U2 - Beta(1, 1)	1.9	4.8	1.3	0.4	0	-
U3 - Beta(0.3, 0.7)	4.9	4.2	0.3	0	0	-
U3 - Beta(0.5, 0.5)	2.1	4.8	1.2	0.3	0	-
U3 - Beta(0.7, 0.3)	1.9	4.4	1.4	0.5	0.1	-
U3 - Beta(1, 1)	2.1	4.9	1.1	0.3	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1137.2
U2 - Beta(0.5, 0.5)						1190.1
U2 - Beta(0.7, 0.3)						1197.6
U2 - Beta(1, 1)						1190.5
U3 - Beta(0.3, 0.7)						1133.1
U3 - Beta(0.5, 0.5)						1183.2
U3 - Beta(0.7, 0.3)						1192.2
U3 - Beta(1, 1)						1181.5
Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	4.6	7.1	8.7	22.2	57.4	0
U2 - Beta(0.5, 0.5)	10.6	10.8	8.3	15	55.3	0
U2 - Beta(0.7, 0.3)	12.3	11.4	10	13.7	52.6	0
U2 - Beta(1, 1)	13	10	8.2	13.8	55	0
U3 - Beta(0.3, 0.7)	9.8	9.9	11.8	21.7	46.8	0
U3 - Beta(0.5, 0.5)	16.5	14.6	9.8	11.1	48	0
U3 - Beta(0.7, 0.3)	19.6	14.4	11.7	10.1	44.2	0
U3 - Beta(1, 1)	21.7	15.4	8.5	10.1	44.3	0
Mean administrations						
U2 - Beta(0.3, 0.7)	3.8	4.7	5.9	7.7	7.8	-
U2 - Beta(0.5, 0.5)	3.4	4	5.1	6.8	10.6	-
U2 - Beta(0.7, 0.3)	3.5	4	5.1	6.6	10.7	-
U2 - Beta(1, 1)	3.6	4.2	5.1	6.6	10.5	-
U3 - Beta(0.3, 0.7)	3.9	4.9	6.3	7.8	7.1	-
U3 - Beta(0.5, 0.5)	3.5	4.2	5.2	6.8	10.3	-
U3 - Beta(0.7, 0.3)	3.7	4.3	5.2	6.5	10.2	-
U3 - Beta(1, 1)	3.8	4.4	5.2	6.6	10	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.5	2.9	2.4	3.1	2.4	-
U2 - Beta(0.5, 0.5)	0.2	2.8	1.8	2.8	3	-
U2 - Beta(0.7, 0.3)	0.2	2.8	1.8	2.8	3	-
U2 - Beta(1, 1)	0.3	2.9	1.8	2.8	3	-
U3 - Beta(0.3, 0.7)	0.5	3	2.5	3.1	2.1	-
U3 - Beta(0.5, 0.5)	0.2	2.9	1.8	2.9	2.9	-
U3 - Beta(0.7, 0.3)	0.3	3	1.8	2.7	2.9	-
U3 - Beta(1, 1)	0.4	3	1.8	2.8	2.8	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1034.8
U2 - Beta(0.5, 0.5)						1067
U2 - Beta(0.7, 0.3)						1068.5
U2 - Beta(1, 1)						1064.9
U3 - Beta(0.3, 0.7)						1028.5
U3 - Beta(0.5, 0.5)						1062.6
U3 - Beta(0.7, 0.3)						1062.1
U3 - Beta(1, 1)						1058.1

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.6	56.4	20.9	1.7	0	12.4
U2 - Beta(0.5, 0.5)	11	53.4	19.8	3	0.1	12.7
U2 - Beta(0.7, 0.3)	11.7	51.9	19.2	3.7	0	13.5
U2 - Beta(1, 1)	11.2	54.1	18.9	3.3	0	12.5
U3 - Beta(0.3, 0.7)	10.8	60.8	15.4	1.2	0	11.8
U3 - Beta(0.5, 0.5)	13.5	55.2	16.6	2.2	0.1	12.4
U3 - Beta(0.7, 0.3)	12.3	55.5	17.4	2.4	0	12.4
U3 - Beta(1, 1)	13.7	54.3	16.9	3	0.1	12
Mean administrations						
U2 - Beta(0.3, 0.7)	7.3	14.8	5.3	1.1	0.1	-
U2 - Beta(0.5, 0.5)	7.1	12.1	6.6	2.3	0.4	-
U2 - Beta(0.7, 0.3)	7.2	11.7	6.5	2.6	0.5	-
U2 - Beta(1, 1)	7.2	12.1	6.6	2.3	0.4	-
U3 - Beta(0.3, 0.7)	7.8	15.4	4.6	0.9	0.1	-
U3 - Beta(0.5, 0.5)	7.4	12.5	6.3	2.1	0.4	-
U3 - Beta(0.7, 0.3)	7.4	12.3	6.1	2.4	0.4	-
U3 - Beta(1, 1)	7.5	12.6	6.1	2.1	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.6	6	1.7	0.4	0	-
U2 - Beta(0.5, 0.5)	1.5	5.2	1.9	0.8	0.1	-
U2 - Beta(0.7, 0.3)	1.5	5.1	1.8	0.9	0.1	-
U2 - Beta(1, 1)	1.4	5.3	1.9	0.8	0.1	-
U3 - Beta(0.3, 0.7)	1.9	6.3	1.5	0.3	0	-
U3 - Beta(0.5, 0.5)	1.5	5.4	1.8	0.7	0.1	-
U3 - Beta(0.7, 0.3)	1.6	5.3	1.7	0.8	0.1	-
U3 - Beta(1, 1)	1.6	5.5	1.7	0.7	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1030.4
U2 - Beta(0.5, 0.5)						1047.1
U2 - Beta(0.7, 0.3)						1051.1
U2 - Beta(1, 1)						1049.7
U3 - Beta(0.3, 0.7)						1026.4
U3 - Beta(0.5, 0.5)						1047.4
U3 - Beta(0.7, 0.3)						1048.9
U3 - Beta(1, 1)						1043.8
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	5.7	44.6	30.5	14.5	2	2.7
U2 - Beta(0.5, 0.5)	10.6	45.6	27.1	12.6	1.9	2.2
U2 - Beta(0.7, 0.3)	9.3	45.3	28.2	13.7	1.6	1.9
U2 - Beta(1, 1)	11.8	43.5	27.5	13.3	1.8	2.1
U3 - Beta(0.3, 0.7)	7.5	50.5	27.1	11.2	1.1	2.6
U3 - Beta(0.5, 0.5)	12.4	50.1	22.6	11.5	1.3	2.1
U3 - Beta(0.7, 0.3)	14.7	46.8	24.1	11.1	1.4	1.9
U3 - Beta(1, 1)	16.4	49	20.2	11.1	1.2	2.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.7	12.5	7.9	3.7	0.9	-
U2 - Beta(0.5, 0.5)	4.6	9.9	8.1	5.2	1.9	-
U2 - Beta(0.7, 0.3)	4.8	9.6	8	5.3	1.9	-
U2 - Beta(1, 1)	4.7	9.8	8.2	5.2	1.8	-
U3 - Beta(0.3, 0.7)	5.1	13.5	7.4	3.1	0.6	-
U3 - Beta(0.5, 0.5)	4.9	10.4	7.8	4.9	1.7	-
U3 - Beta(0.7, 0.3)	5	10.1	7.7	5.1	1.8	-
U3 - Beta(1, 1)	5.1	10.5	7.6	4.8	1.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.8	5.6	2.7	1.3	0.3	-
U2 - Beta(0.5, 0.5)	0.6	4.8	2.6	1.8	0.6	-
U2 - Beta(0.7, 0.3)	0.7	4.8	2.4	1.9	0.6	-
U2 - Beta(1, 1)	0.7	4.8	2.6	1.8	0.6	-
U3 - Beta(0.3, 0.7)	0.9	6	2.6	1.1	0.2	-
U3 - Beta(0.5, 0.5)	0.8	5	2.5	1.7	0.6	-
U3 - Beta(0.7, 0.3)	0.8	4.9	2.4	1.8	0.6	-
U3 - Beta(1, 1)	0.9	5	2.4	1.7	0.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1040
U2 - Beta(0.5, 0.5)						1059.4
U2 - Beta(0.7, 0.3)						1059.7
U2 - Beta(1, 1)						1059.1
U3 - Beta(0.3, 0.7)						1034.3
U3 - Beta(0.5, 0.5)						1056.2
U3 - Beta(0.7, 0.3)						1057.2
U3 - Beta(1, 1)						1054.2

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	64	0.7	0	0	0	35.3
U2 - Beta(0.5, 0.5)	58.8	1.8	0	0	0	39.4
U2 - Beta(0.7, 0.3)	57.9	1.7	0	0	0	40.4
U2 - Beta(1, 1)	58.9	1.9	0	0	0	39.2
U3 - Beta(0.3, 0.7)	64	0.6	0	0	0	35.4
U3 - Beta(0.5, 0.5)	59	1.6	0	0	0	39.4
U3 - Beta(0.7, 0.3)	58.8	2	0	0	0	39.2
U3 - Beta(1, 1)	59.2	1.8	0	0	0	39
Mean administrations						
U2 - Beta(0.3, 0.7)	23.1	0.7	0	0	0	-
U2 - Beta(0.5, 0.5)	20.3	2.7	0.3	0	0	-
U2 - Beta(0.7, 0.3)	19.1	3.6	0.4	0.1	0	-
U2 - Beta(1, 1)	20.4	2.6	0.2	0	0	-
U3 - Beta(0.3, 0.7)	23.1	0.7	0	0	0	-
U3 - Beta(0.5, 0.5)	20.4	2.6	0.2	0	0	-
U3 - Beta(0.7, 0.3)	19.8	3.2	0.3	0	0	-
U3 - Beta(1, 1)	20.5	2.5	0.2	0	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	4.9	0.3	0	0	0	-
U2 - Beta(0.5, 0.5)	3.8	1	0.1	0	0	-
U2 - Beta(0.7, 0.3)	3.5	1.2	0.1	0	0	-
U2 - Beta(1, 1)	3.9	1	0.1	0	0	-
U3 - Beta(0.3, 0.7)	4.9	0.3	0	0	0	-
U3 - Beta(0.5, 0.5)	3.9	1	0.1	0	0	-
U3 - Beta(0.7, 0.3)	3.6	1.2	0.1	0	0	-
U3 - Beta(1, 1)	3.9	1	0	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1032
U2 - Beta(0.5, 0.5)						1012.1
U2 - Beta(0.7, 0.3)						1006.2
U2 - Beta(1, 1)						1012.6
U3 - Beta(0.3, 0.7)						1032.3
U3 - Beta(0.5, 0.5)						1009.4
U3 - Beta(0.7, 0.3)						1013.9
U3 - Beta(1, 1)						1013.3
Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.3	58.4	14.7	2.9	0.3	10.4
U2 - Beta(0.5, 0.5)	14.4	56.9	13.6	4	0	11.1
U2 - Beta(0.7, 0.3)	14.1	56	14.3	4.2	0.1	11.3
U2 - Beta(1, 1)	14.7	56.6	13.3	4.3	0	11.1
U3 - Beta(0.3, 0.7)	15.2	59.3	12	2.6	0.1	10.8
U3 - Beta(0.5, 0.5)	16.5	57.2	11.8	3.2	0	11.3
U3 - Beta(0.7, 0.3)	17	55.5	12.1	4.1	0	11.3
U3 - Beta(1, 1)	18.1	55	11.3	4.4	0	11.2
Mean administrations						
U2 - Beta(0.3, 0.7)	7.7	14.5	5.3	1.3	0.1	-
U2 - Beta(0.5, 0.5)	7.8	12.9	5.8	2.2	0.2	-
U2 - Beta(0.7, 0.3)	7.9	12.5	5.8	2.4	0.4	-
U2 - Beta(1, 1)	7.8	12.8	5.8	2.3	0.2	-
U3 - Beta(0.3, 0.7)	8.2	14.9	4.7	1.1	0	-
U3 - Beta(0.5, 0.5)	8.2	13	5.5	2.1	0.2	-
U3 - Beta(0.7, 0.3)	8.2	12.7	5.5	2.2	0.2	-
U3 - Beta(1, 1)	8.3	12.9	5.5	2.1	0.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.9	6	1.8	0.4	0	-
U2 - Beta(0.5, 0.5)	1.8	5.7	1.6	0.7	0.1	-
U2 - Beta(0.7, 0.3)	1.8	5.6	1.6	0.8	0.1	-
U2 - Beta(1, 1)	1.8	5.6	1.7	0.8	0.1	-
U3 - Beta(0.3, 0.7)	2.1	6.2	1.6	0.3	0	-
U3 - Beta(0.5, 0.5)	2	5.7	1.6	0.7	0.1	-
U3 - Beta(0.7, 0.3)	2	5.6	1.5	0.7	0.1	-
U3 - Beta(1, 1)	2	5.7	1.5	0.7	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1033.7
U2 - Beta(0.5, 0.5)						1045.6
U2 - Beta(0.7, 0.3)						1048.8
U2 - Beta(1, 1)						1046
U3 - Beta(0.3, 0.7)						1031
U3 - Beta(0.5, 0.5)						1039.7
U3 - Beta(0.7, 0.3)						1043.3
U3 - Beta(1, 1)						1041.2

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.3	1.5	0.2	0	0	90
U2 - Beta(0.5, 0.5)	8.1	2	0	0	0	89.9
U2 - Beta(0.7, 0.3)	8.3	2.1	0	0	0	89.6
U2 - Beta(1, 1)	8.1	2.6	0	0	0	89.3
U3 - Beta(0.3, 0.7)	7.7	1.6	0	0	0	90.7
U3 - Beta(0.5, 0.5)	8.3	1.7	0.1	0	0	89.9
U3 - Beta(0.7, 0.3)	8.2	1.9	0.1	0	0	89.8
U3 - Beta(1, 1)	8.4	2.2	0.2	0	0	89.2
Mean administrations						
U2 - Beta(0.3, 0.7)	11.3	4.2	0.7	0.1	0	-
U2 - Beta(0.5, 0.5)	11.1	4.4	0.7	0.1	0	-
U2 - Beta(0.7, 0.3)	11.1	4.4	0.7	0.1	0	-
U2 - Beta(1, 1)	11.1	4.5	0.7	0.1	0	-
U3 - Beta(0.3, 0.7)	11.3	4.1	0.7	0.1	0	-
U3 - Beta(0.5, 0.5)	11.1	4.4	0.7	0.1	0	-
U3 - Beta(0.7, 0.3)	11.1	4.4	0.7	0.1	0	-
U3 - Beta(1, 1)	11.1	4.4	0.7	0.1	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.9	2.2	0.2	0	0	-
U2 - Beta(0.5, 0.5)	2.9	2.3	0.2	0	0	-
U2 - Beta(0.7, 0.3)	2.8	2.3	0.2	0	0	-
U2 - Beta(1, 1)	2.8	2.3	0.2	0	0	-
U3 - Beta(0.3, 0.7)	3	2.1	0.2	0	0	-
U3 - Beta(0.5, 0.5)	2.8	2.3	0.2	0	0	-
U3 - Beta(0.7, 0.3)	2.8	2.3	0.2	0	0	-
U3 - Beta(1, 1)	2.8	2.3	0.2	0	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						594.9
U2 - Beta(0.5, 0.5)						594.5
U2 - Beta(0.7, 0.3)						593.8
U2 - Beta(1, 1)						594.7
U3 - Beta(0.3, 0.7)						590.7
U3 - Beta(0.5, 0.5)						591.5
U3 - Beta(0.7, 0.3)						592.6
U3 - Beta(1, 1)						594.5
Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	0.2	0.9	7.6	72.2	19.1	0
U2 - Beta(0.5, 0.5)	2.2	2.2	3.6	54.1	37.9	0
U2 - Beta(0.7, 0.3)	3.9	3.1	6	49.3	37.7	0
U2 - Beta(1, 1)	3.1	2.3	4.1	53.8	36.7	0
U3 - Beta(0.3, 0.7)	0.8	1.3	9.7	74.1	14.1	0
U3 - Beta(0.5, 0.5)	4	3.1	4.8	57.1	31	0
U3 - Beta(0.7, 0.3)	5.5	5.1	6.5	51.3	31.6	0
U3 - Beta(1, 1)	4.9	4.4	5.7	55.5	29.5	0
Mean administrations						
U2 - Beta(0.3, 0.7)	3.6	4	6.5	13.5	2.4	-
U2 - Beta(0.5, 0.5)	3.1	3.1	3.9	12.5	7.4	-
U2 - Beta(0.7, 0.3)	3.1	3.1	3.9	10.9	9	-
U2 - Beta(1, 1)	3.1	3.2	3.9	12.5	7.2	-
U3 - Beta(0.3, 0.7)	3.6	4.1	6.8	13.6	2	-
U3 - Beta(0.5, 0.5)	3.1	3.2	4	13	6.7	-
U3 - Beta(0.7, 0.3)	3.1	3.2	4.1	11.4	8.2	-
U3 - Beta(1, 1)	3.1	3.2	4.1	13	6.6	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	2.6	2.5	4.6	0.8	-
U2 - Beta(0.5, 0.5)	0	2.4	1.2	4.4	2.2	-
U2 - Beta(0.7, 0.3)	0	2.4	1.2	3.9	2.5	-
U2 - Beta(1, 1)	0	2.5	1.2	4.4	2.1	-
U3 - Beta(0.3, 0.7)	0.4	2.7	2.7	4.7	0.6	-
U3 - Beta(0.5, 0.5)	0	2.5	1.3	4.6	2	-
U3 - Beta(0.7, 0.3)	0.1	2.5	1.3	4.1	2.3	-
U3 - Beta(1, 1)	0	2.5	1.3	4.6	1.9	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1052.3
U2 - Beta(0.5, 0.5)						1090.7
U2 - Beta(0.7, 0.3)						1093.8
U2 - Beta(1, 1)						1089.1
U3 - Beta(0.3, 0.7)						1048.3
U3 - Beta(0.5, 0.5)						1087.4
U3 - Beta(0.7, 0.3)						1090
U3 - Beta(1, 1)						1088.4

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	22	33.3	41.1	1.9	0.4	1.3
U2 - Beta(0.5, 0.5)	25.6	29.7	37.1	6.3	0.1	1.2
U2 - Beta(0.7, 0.3)	26.9	30	35.1	6.6	0.1	1.3
U2 - Beta(1, 1)	26.6	29.7	36.1	6.1	0.1	1.4
U3 - Beta(0.3, 0.7)	29.9	37.1	30.4	1.6	0.1	0.9
U3 - Beta(0.5, 0.5)	31.9	29.3	31.9	5.6	0	1.3
U3 - Beta(0.7, 0.3)	30.5	29.5	32.5	6	0.2	1.3
U3 - Beta(1, 1)	31.7	30	30.6	6.4	0	1.3
Mean administrations						
U2 - Beta(0.3, 0.7)	9.9	10.4	8.4	1.1	0.1	-
U2 - Beta(0.5, 0.5)	8.7	9	8.8	2.9	0.4	-
U2 - Beta(0.7, 0.3)	9	9	8	3.3	0.6	-
U2 - Beta(1, 1)	8.6	9.1	8.9	2.9	0.4	-
U3 - Beta(0.3, 0.7)	11.3	11	6.8	0.8	0.1	-
U3 - Beta(0.5, 0.5)	9.4	9.1	8.5	2.5	0.3	-
U3 - Beta(0.7, 0.3)	9.3	9	8	3.1	0.4	-
U3 - Beta(1, 1)	9.5	9.3	8.3	2.5	0.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2.8	4.3	2.6	0.4	0	-
U2 - Beta(0.5, 0.5)	2.2	4	2.4	0.9	0.1	-
U2 - Beta(0.7, 0.3)	2.3	4	2	1	0.1	-
U2 - Beta(1, 1)	2.2	4	2.4	0.9	0.1	-
U3 - Beta(0.3, 0.7)	3.4	4.6	2.1	0.3	0	-
U3 - Beta(0.5, 0.5)	2.5	4	2.3	0.8	0.1	-
U3 - Beta(0.7, 0.3)	2.5	4	2.1	1	0.1	-
U3 - Beta(1, 1)	2.5	4.1	2.2	0.8	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1085.4
U2 - Beta(0.5, 0.5)						1121.3
U2 - Beta(0.7, 0.3)						1123.8
U2 - Beta(1, 1)						1122.2
U3 - Beta(0.3, 0.7)						1073.3
U3 - Beta(0.5, 0.5)						1114.8
U3 - Beta(0.7, 0.3)						1120.5
U3 - Beta(1, 1)						1113.9

Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	13.9	26.8	43.4	4.8	0.2	10.9
U2 - Beta(0.5, 0.5)	18.7	29.8	32.9	7.8	0.1	10.7
U2 - Beta(0.7, 0.3)	18.8	29.6	31.1	9.6	0.1	10.8
U2 - Beta(1, 1)	17.8	28.9	32.7	9.6	0.1	10.9
U3 - Beta(0.3, 0.7)	19.3	29.4	35.8	4.8	0.1	10.6
U3 - Beta(0.5, 0.5)	21.6	28.7	31	7.3	0.1	11.3
U3 - Beta(0.7, 0.3)	24.8	28.4	27.5	8.1	0.1	11.1
U3 - Beta(1, 1)	24.2	28.6	27.9	7.9	0.1	11.3
Mean administrations						
U2 - Beta(0.3, 0.7)	8.1	9.5	9.1	1.8	0.1	-
U2 - Beta(0.5, 0.5)	8	8.6	8.3	3.2	0.4	-
U2 - Beta(0.7, 0.3)	8.2	8.8	7.8	3.4	0.5	-
U2 - Beta(1, 1)	7.8	8.8	8.4	3.2	0.5	-
U3 - Beta(0.3, 0.7)	9	9.8	8.2	1.4	0.1	-
U3 - Beta(0.5, 0.5)	8.3	8.7	8.2	3	0.4	-
U3 - Beta(0.7, 0.3)	8.4	8.8	7.7	3.3	0.4	-
U3 - Beta(1, 1)	8.5	8.9	7.9	3	0.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2	4.3	2.9	0.7	0	-
U2 - Beta(0.5, 0.5)	1.9	4.1	2.3	1.1	0.1	-
U2 - Beta(0.7, 0.3)	2	4.2	2.1	1.1	0.1	-
U2 - Beta(1, 1)	1.8	4.2	2.3	1.1	0.1	-
U3 - Beta(0.3, 0.7)	2.4	4.4	2.6	0.6	0	-
U3 - Beta(0.5, 0.5)	2	4.2	2.3	1	0.1	-
U3 - Beta(0.7, 0.3)	2.1	4.2	2.1	1.1	0.1	-
U3 - Beta(1, 1)	2.1	4.3	2.2	1.1	0.1	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1024.7
U2 - Beta(0.5, 0.5)						1045.7
U2 - Beta(0.7, 0.3)						1048.4
U2 - Beta(1, 1)						1047
U3 - Beta(0.3, 0.7)						1016.4
U3 - Beta(0.5, 0.5)						1041.9
U3 - Beta(0.7, 0.3)						1043
U3 - Beta(1, 1)						1040.9

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.1	22.7	56.5	7.7	0.4	4.6
U2 - Beta(0.5, 0.5)	12.9	23.4	46.3	11.8	0.7	4.9
U2 - Beta(0.7, 0.3)	14.3	22.1	44.8	12.9	1.3	4.6
U2 - Beta(1, 1)	13.4	23.6	45.2	12.3	0.8	4.7
U3 - Beta(0.3, 0.7)	13.3	26.4	49.1	6.2	0.3	4.7
U3 - Beta(0.5, 0.5)	15.8	25.7	42.3	11	0.5	4.7
U3 - Beta(0.7, 0.3)	18.1	25.3	40.3	11.1	0.7	4.5
U3 - Beta(1, 1)	18.5	24.4	41.7	10.4	0.5	4.5
Mean administrations						
U2 - Beta(0.3, 0.7)	6.6	9.3	11.2	2.1	0.2	-
U2 - Beta(0.5, 0.5)	6.2	8	10.3	4.1	0.9	-
U2 - Beta(0.7, 0.3)	6.4	7.9	9.5	4.5	1.1	-
U2 - Beta(1, 1)	6.2	8.1	10.2	4.1	0.9	-
U3 - Beta(0.3, 0.7)	7.4	9.7	10.4	1.7	0.2	-
U3 - Beta(0.5, 0.5)	6.5	8.2	10.1	3.9	0.7	-
U3 - Beta(0.7, 0.3)	6.6	8.2	9.4	4.4	0.8	-
U3 - Beta(1, 1)	6.6	8.5	9.8	3.8	0.7	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.5	4.2	3.6	0.7	0.1	-
U2 - Beta(0.5, 0.5)	1.2	3.9	2.9	1.3	0.3	-
U2 - Beta(0.7, 0.3)	1.4	3.9	2.6	1.4	0.3	-
U2 - Beta(1, 1)	1.2	3.9	2.9	1.3	0.3	-
U3 - Beta(0.3, 0.7)	1.9	4.4	3.3	0.6	0	-
U3 - Beta(0.5, 0.5)	1.4	4	2.9	1.3	0.2	-
U3 - Beta(0.7, 0.3)	1.4	4	2.6	1.4	0.2	-
U3 - Beta(1, 1)	1.4	4.1	2.8	1.3	0.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1062.9
U2 - Beta(0.5, 0.5)						1087.9
U2 - Beta(0.7, 0.3)						1090.3
U2 - Beta(1, 1)						1088.4
U3 - Beta(0.3, 0.7)						1053.5
U3 - Beta(0.5, 0.5)						1085.5
U3 - Beta(0.7, 0.3)						1088.5
U3 - Beta(1, 1)						1082.9
Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	8.5	47.9	7.3	0.4	0	35.9
U2 - Beta(0.5, 0.5)	9	44.6	9.1	0.6	0	36.7
U2 - Beta(0.7, 0.3)	8.8	43.7	9.2	0.7	0	37.6
U2 - Beta(1, 1)	10.2	43.9	8.6	0.5	0	36.8
U3 - Beta(0.3, 0.7)	8.4	49.6	5.3	0.3	0	36.4
U3 - Beta(0.5, 0.5)	8.9	45.4	7.9	0.1	0	37.7
U3 - Beta(0.7, 0.3)	9.2	46.2	8	0.6	0	36
U3 - Beta(1, 1)	9.4	45.4	8.5	0.5	0	36.2
Mean administrations						
U2 - Beta(0.3, 0.7)	8.5	12.9	3.1	0.5	0	-
U2 - Beta(0.5, 0.5)	8.5	11.2	4.1	1.2	0.2	-
U2 - Beta(0.7, 0.3)	8.5	10.9	4.2	1.5	0.2	-
U2 - Beta(1, 1)	8.5	11.1	4.1	1.2	0.2	-
U3 - Beta(0.3, 0.7)	8.7	13.2	2.6	0.4	0	-
U3 - Beta(0.5, 0.5)	8.7	11.2	4	1.1	0.1	-
U3 - Beta(0.7, 0.3)	8.7	11	4	1.3	0.2	-
U3 - Beta(1, 1)	8.7	11.3	3.9	1.1	0.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.9	5.3	1	0.2	0	-
U2 - Beta(0.5, 0.5)	1.9	4.8	1.2	0.4	0	-
U2 - Beta(0.7, 0.3)	1.9	4.7	1.2	0.5	0.1	-
U2 - Beta(1, 1)	1.9	4.7	1.2	0.4	0.1	-
U3 - Beta(0.3, 0.7)	2	5.5	0.8	0.1	0	-
U3 - Beta(0.5, 0.5)	2	4.8	1.2	0.4	0	-
U3 - Beta(0.7, 0.3)	1.9	4.7	1.1	0.4	0.1	-
U3 - Beta(1, 1)	1.9	4.8	1.1	0.4	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						911
U2 - Beta(0.5, 0.5)						928
U2 - Beta(0.7, 0.3)						928.2
U2 - Beta(1, 1)						924.8
U3 - Beta(0.3, 0.7)						905.2
U3 - Beta(0.5, 0.5)						921.7
U3 - Beta(0.7, 0.3)						926.7
U3 - Beta(1, 1)						923

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	73.6	21.1	2.9	0	0	2.4
U2 - Beta(0.5, 0.5)	63.1	30	4.4	0.2	0	2.3
U2 - Beta(0.7, 0.3)	62.7	29.8	4.6	0.3	0	2.6
U2 - Beta(1, 1)	62	31.2	4.3	0.1	0	2.4
U3 - Beta(0.3, 0.7)	76.6	19.3	1.8	0	0	2.3
U3 - Beta(0.5, 0.5)	67.2	27.3	3	0.2	0	2.3
U3 - Beta(0.7, 0.3)	65.9	27	4.6	0.2	0	2.3
U3 - Beta(1, 1)	66.6	27.8	3.1	0.1	0	2.4
Mean administrations						
U2 - Beta(0.3, 0.7)	23.3	5.2	0.9	0.1	0	-
U2 - Beta(0.5, 0.5)	17.3	8.9	2.7	0.6	0	-
U2 - Beta(0.7, 0.3)	17	8.8	2.8	0.9	0.1	-
U2 - Beta(1, 1)	17.2	9.1	2.6	0.6	0	-
U3 - Beta(0.3, 0.7)	23.8	5.1	0.6	0.1	0	-
U3 - Beta(0.5, 0.5)	18.2	8.6	2.2	0.5	0	-
U3 - Beta(0.7, 0.3)	17.2	8.9	2.8	0.7	0.1	-
U3 - Beta(1, 1)	18.3	8.6	2.1	0.4	0	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	6.4	1.9	0.3	0	0	-
U2 - Beta(0.5, 0.5)	4.2	3.1	0.7	0.2	0	-
U2 - Beta(0.7, 0.3)	4.2	3.1	0.7	0.2	0	-
U2 - Beta(1, 1)	4.2	3.2	0.7	0.2	0	-
U3 - Beta(0.3, 0.7)	6.6	1.8	0.2	0	0	-
U3 - Beta(0.5, 0.5)	4.6	3	0.6	0.1	0	-
U3 - Beta(0.7, 0.3)	4.2	3.1	0.7	0.2	0	-
U3 - Beta(1, 1)	4.6	3.1	0.5	0.1	0	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1153.7
U2 - Beta(0.5, 0.5)						1177.9
U2 - Beta(0.7, 0.3)						1180.6
U2 - Beta(1, 1)						1177.1
U3 - Beta(0.3, 0.7)						1153.9
U3 - Beta(0.5, 0.5)						1172.2
U3 - Beta(0.7, 0.3)						1179.7
U3 - Beta(1, 1)						1171.8
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.9	9.1	15.7	58	8.9	0.4
U2 - Beta(0.5, 0.5)	17.2	12.2	12.2	43.7	14.3	0.4
U2 - Beta(0.7, 0.3)	20.3	13.6	12.6	39.4	13.7	0.4
U2 - Beta(1, 1)	19.8	11.2	12.4	41.3	14.9	0.4
U3 - Beta(0.3, 0.7)	15.3	13.7	15.6	49.8	5.1	0.5
U3 - Beta(0.5, 0.5)	25.6	14.5	10.1	38.4	10.9	0.5
U3 - Beta(0.7, 0.3)	28.1	16	10.8	33.7	11	0.4
U3 - Beta(1, 1)	27.6	16.7	10.3	34.9	10.1	0.4
Mean administrations						
U2 - Beta(0.3, 0.7)	5.5	6	7.1	9.6	1.6	-
U2 - Beta(0.5, 0.5)	4.3	4.8	5.9	10.4	4.6	-
U2 - Beta(0.7, 0.3)	4.5	4.9	6	9.5	5.1	-
U2 - Beta(1, 1)	4.3	4.9	5.9	10.2	4.7	-
U3 - Beta(0.3, 0.7)	6.2	6.5	7.1	8.8	1.2	-
U3 - Beta(0.5, 0.5)	4.5	5	6	10.3	4.3	-
U3 - Beta(0.7, 0.3)	4.8	5.1	5.9	9.6	4.6	-
U3 - Beta(1, 1)	4.7	5.1	5.8	10.2	4.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.2	3.2	2.7	3.2	0.6	-
U2 - Beta(0.5, 0.5)	0.6	2.9	1.9	3.4	1.3	-
U2 - Beta(0.7, 0.3)	0.6	3	1.8	3.2	1.4	-
U2 - Beta(1, 1)	0.5	3	1.9	3.3	1.3	-
U3 - Beta(0.3, 0.7)	1.5	3.4	2.7	3	0.4	-
U3 - Beta(0.5, 0.5)	0.6	3	1.9	3.4	1.2	-
U3 - Beta(0.7, 0.3)	0.8	3.1	1.8	3.1	1.3	-
U3 - Beta(1, 1)	0.7	3.1	1.8	3.3	1.2	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1051.2
U2 - Beta(0.5, 0.5)						1096.3
U2 - Beta(0.7, 0.3)						1096.6
U2 - Beta(1, 1)						1094.5
U3 - Beta(0.3, 0.7)						1041.2
U3 - Beta(0.5, 0.5)						1091.3
U3 - Beta(0.7, 0.3)						1089.9
U3 - Beta(1, 1)						1088.1

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	4.6	72.1	19.1	4.2	0	0
U2 - Beta(0.5, 0.5)	5.3	34.7	27	25.9	7.1	0
U2 - Beta(0.7, 0.3)	8.8	34.4	26.3	22.6	7.9	0
U2 - Beta(1, 1)	6	36.6	26.7	24	6.7	0
U3 - Beta(0.3, 0.7)	6.4	74.6	15.2	3.6	0.2	0
U3 - Beta(0.5, 0.5)	6.8	41.2	26.4	21.2	4.4	0
U3 - Beta(0.7, 0.3)	9	39.6	26.4	20.7	4.3	0
U3 - Beta(1, 1)	7.5	42.2	26.1	20.5	3.7	0
Mean administrations						
U2 - Beta(0.3, 0.7)	7.5	18.1	3.8	0.6	0.1	-
U2 - Beta(0.5, 0.5)	3.8	11	7.4	5.9	2	-
U2 - Beta(0.7, 0.3)	3.6	6.4	7.5	8.3	4.2	-
U2 - Beta(1, 1)	3.7	11.1	7.4	5.8	2	-
U3 - Beta(0.3, 0.7)	7.8	18.4	3.2	0.6	0	-
U3 - Beta(0.5, 0.5)	3.9	11.9	7.2	5.4	1.6	-
U3 - Beta(0.7, 0.3)	3.8	8.1	7.6	7.4	3.1	-
U3 - Beta(1, 1)	3.9	12.1	7.2	5.3	1.5	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2	6.5	1.2	0.2	0	-
U2 - Beta(0.5, 0.5)	0.3	4.4	2.2	1.7	0.6	-
U2 - Beta(0.7, 0.3)	0.2	3	2.1	2.2	1.1	-
U2 - Beta(1, 1)	0.2	4.4	2.2	1.7	0.6	-
U3 - Beta(0.3, 0.7)	2.1	6.6	1	0.2	0	-
U3 - Beta(0.5, 0.5)	0.3	4.7	2.1	1.6	0.5	-
U3 - Beta(0.7, 0.3)	0.3	3.5	2.2	2.1	0.8	-
U3 - Beta(1, 1)	0.3	4.8	2.1	1.5	0.5	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1108.4
U2 - Beta(0.5, 0.5)						1153.1
U2 - Beta(0.7, 0.3)						1178.7
U2 - Beta(1, 1)						1152.8
U3 - Beta(0.3, 0.7)						1107.3
U3 - Beta(0.5, 0.5)						1147.9
U3 - Beta(0.7, 0.3)						1164.2
U3 - Beta(1, 1)						1146.2
Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.8	7.4	41.2	35.2	12.3	0.1
U2 - Beta(0.5, 0.5)	13.6	14.2	34.4	25.7	12	0.1
U2 - Beta(0.7, 0.3)	15.1	14.7	32.6	25.7	11.8	0.1
U2 - Beta(1, 1)	16.1	14.7	31.4	26	11.7	0.1
U3 - Beta(0.3, 0.7)	8.9	10.8	42.5	30.6	7.1	0.1
U3 - Beta(0.5, 0.5)	20	16.5	32.5	23.6	7.3	0.1
U3 - Beta(0.7, 0.3)	20.6	16.9	33.1	20.6	8.7	0.1
U3 - Beta(1, 1)	23.6	17.7	30.3	21.4	6.9	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	4.2	5.7	10.9	6.9	2.2	-
U2 - Beta(0.5, 0.5)	3.5	4.3	8.5	8.4	5.3	-
U2 - Beta(0.7, 0.3)	3.6	4.4	8.4	8.3	5.4	-
U2 - Beta(1, 1)	3.6	4.4	8.5	8.4	5.1	-
U3 - Beta(0.3, 0.7)	4.4	6	11.3	6.6	1.8	-
U3 - Beta(0.5, 0.5)	3.6	4.5	8.8	8.4	4.7	-
U3 - Beta(0.7, 0.3)	3.7	4.7	8.7	8.1	4.8	-
U3 - Beta(1, 1)	3.9	4.7	8.6	8.4	4.4	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.6	3.2	4.1	2.5	0.8	-
U2 - Beta(0.5, 0.5)	0.2	2.9	2.8	3.2	1.7	-
U2 - Beta(0.7, 0.3)	0.3	2.9	2.7	3.2	1.7	-
U2 - Beta(1, 1)	0.2	2.9	2.8	3.2	1.7	-
U3 - Beta(0.3, 0.7)	0.7	3.3	4.2	2.4	0.6	-
U3 - Beta(0.5, 0.5)	0.3	2.9	2.9	3.2	1.5	-
U3 - Beta(0.7, 0.3)	0.3	3	2.8	3.1	1.5	-
U3 - Beta(1, 1)	0.4	3	2.8	3.2	1.4	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1033.3
U2 - Beta(0.5, 0.5)						1057.6
U2 - Beta(0.7, 0.3)						1061.7
U2 - Beta(1, 1)						1056.3
U3 - Beta(0.3, 0.7)						1029.4
U3 - Beta(0.5, 0.5)						1054.2
U3 - Beta(0.7, 0.3)						1057.4
U3 - Beta(1, 1)						1052.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	3.3	3.6	12.2	60.7	20.1	0.1
U2 - Beta(0.5, 0.5)	7.9	5.6	12.1	43.7	30.6	0.1
U2 - Beta(0.7, 0.3)	9.6	7.2	11.5	42	29.5	0.2
U2 - Beta(1, 1)	9.5	6.4	10.4	43	30.5	0.2
U3 - Beta(0.3, 0.7)	6.7	4.5	14.8	59.9	14	0.1
U3 - Beta(0.5, 0.5)	12	7.6	11	43.6	25.7	0.1
U3 - Beta(0.7, 0.3)	14.7	9.2	11.4	40.6	23.9	0.2
U3 - Beta(1, 1)	16.9	8.2	10.8	40.6	23.4	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	3.8	4.2	7.5	11.6	2.9	-
U2 - Beta(0.5, 0.5)	3.3	3.6	5.9	10.5	6.7	-
U2 - Beta(0.7, 0.3)	3.3	3.7	5.8	9.7	7.6	-
U2 - Beta(1, 1)	3.4	3.7	5.9	10.3	6.7	-
U3 - Beta(0.3, 0.7)	3.8	4.3	8	11.6	2.3	-
U3 - Beta(0.5, 0.5)	3.4	3.8	6	10.6	6.2	-
U3 - Beta(0.7, 0.3)	3.6	3.9	6	10	6.6	-
U3 - Beta(1, 1)	3.6	3.9	5.9	10.5	6.1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.4	2.7	2.8	3.9	0.9	-
U2 - Beta(0.5, 0.5)	0.1	2.7	1.8	3.7	1.8	-
U2 - Beta(0.7, 0.3)	0.1	2.7	1.7	3.6	1.9	-
U2 - Beta(1, 1)	0.2	2.7	1.8	3.7	1.8	-
U3 - Beta(0.3, 0.7)	0.4	2.8	3	3.9	0.7	-
U3 - Beta(0.5, 0.5)	0.2	2.7	1.8	3.8	1.6	-
U3 - Beta(0.7, 0.3)	0.3	2.8	1.8	3.6	1.7	-
U3 - Beta(1, 1)	0.3	2.8	1.8	3.7	1.6	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1061.2
U2 - Beta(0.5, 0.5)						1094.7
U2 - Beta(0.7, 0.3)						1098.6
U2 - Beta(1, 1)						1092.6
U3 - Beta(0.3, 0.7)						1055
U3 - Beta(0.5, 0.5)						1090.6
U3 - Beta(0.7, 0.3)						1089.2
U3 - Beta(1, 1)						1088.6
Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	7.9	9.3	53.4	21.1	4.3	4
U2 - Beta(0.5, 0.5)	11.5	12.3	44.5	23	5.1	3.6
U2 - Beta(0.7, 0.3)	13.4	12	42.2	23.7	5.6	3.1
U2 - Beta(1, 1)	14.2	12.4	42.3	22.2	5.3	3.6
U3 - Beta(0.3, 0.7)	11.5	12.8	49.3	19.4	2.8	4.2
U3 - Beta(0.5, 0.5)	15.8	14.3	42.1	20.9	3.3	3.6
U3 - Beta(0.7, 0.3)	18	14.6	38.7	22	3.6	3.1
U3 - Beta(1, 1)	18.6	14.4	38.5	21.6	3.5	3.4
Mean administrations						
U2 - Beta(0.3, 0.7)	5	6.4	12.4	4.9	1	-
U2 - Beta(0.5, 0.5)	5	6.2	9.6	6.3	2.8	-
U2 - Beta(0.7, 0.3)	4.9	6.2	9.4	6.4	2.9	-
U2 - Beta(1, 1)	5	6.1	9.5	6.4	2.7	-
U3 - Beta(0.3, 0.7)	5.4	6.7	12.1	4.6	0.8	-
U3 - Beta(0.5, 0.5)	5.2	6.3	9.7	6.1	2.4	-
U3 - Beta(0.7, 0.3)	5.4	6.4	9.1	6.3	2.6	-
U3 - Beta(1, 1)	5.6	6.5	9.3	6.1	2.3	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	0.9	3.4	4.1	1.7	0.4	-
U2 - Beta(0.5, 0.5)	0.8	3.5	2.9	2.2	0.9	-
U2 - Beta(0.7, 0.3)	0.8	3.5	2.9	2.2	0.9	-
U2 - Beta(1, 1)	0.9	3.5	2.9	2.2	0.9	-
U3 - Beta(0.3, 0.7)	1.1	3.5	4.1	1.6	0.3	-
U3 - Beta(0.5, 0.5)	1	3.6	2.9	2.1	0.8	-
U3 - Beta(0.7, 0.3)	1.1	3.6	2.8	2.2	0.8	-
U3 - Beta(1, 1)	1.2	3.6	2.8	2.1	0.8	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1053.3
U2 - Beta(0.5, 0.5)						1065.3
U2 - Beta(0.7, 0.3)						1065.1
U2 - Beta(1, 1)						1065.2
U3 - Beta(0.3, 0.7)						1047.7
U3 - Beta(0.5, 0.5)						1062.1
U3 - Beta(0.7, 0.3)						1060.7
U3 - Beta(1, 1)						1057.8

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	5.9	65.8	23.4	4	0.8	0.1
U2 - Beta(0.5, 0.5)	9.2	40	31.6	15.2	3.9	0.1
U2 - Beta(0.7, 0.3)	10.9	39.3	29.8	16.7	3.2	0.1
U2 - Beta(1, 1)	10.3	39.8	30.5	15.6	3.7	0.1
U3 - Beta(0.3, 0.7)	9.1	69.2	18.6	2.8	0.2	0.1
U3 - Beta(0.5, 0.5)	13.6	45.6	27.4	10.6	2.7	0.1
U3 - Beta(0.7, 0.3)	14.2	45.4	25.6	12.9	1.8	0.1
U3 - Beta(1, 1)	14.6	45.5	27	10.1	2.7	0.1
Mean administrations						
U2 - Beta(0.3, 0.7)	7.6	16.9	4.6	0.7	0.1	-
U2 - Beta(0.5, 0.5)	4.4	11.1	8.4	4.5	1.5	-
U2 - Beta(0.7, 0.3)	4.3	8.8	8.6	5.7	2.5	-
U2 - Beta(1, 1)	4.4	11.2	8.4	4.5	1.5	-
U3 - Beta(0.3, 0.7)	8	17.4	3.9	0.6	0.1	-
U3 - Beta(0.5, 0.5)	4.6	12	8.1	4	1.3	-
U3 - Beta(0.7, 0.3)	4.6	10.1	8.3	5.1	1.8	-
U3 - Beta(1, 1)	4.7	12.2	8	3.9	1.2	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	2	6.3	1.4	0.2	0	-
U2 - Beta(0.5, 0.5)	0.5	4.4	2.3	1.3	0.4	-
U2 - Beta(0.7, 0.3)	0.5	3.8	2.2	1.7	0.6	-
U2 - Beta(1, 1)	0.5	4.5	2.3	1.3	0.4	-
U3 - Beta(0.3, 0.7)	2.2	6.4	1.2	0.2	0	-
U3 - Beta(0.5, 0.5)	0.6	4.8	2.2	1.2	0.3	-
U3 - Beta(0.7, 0.3)	0.6	4.2	2.2	1.5	0.5	-
U3 - Beta(1, 1)	0.6	4.9	2.2	1.2	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						1103.3
U2 - Beta(0.5, 0.5)						1154.1
U2 - Beta(0.7, 0.3)						1168
U2 - Beta(1, 1)						1153.4
U3 - Beta(0.3, 0.7)						1099.5
U3 - Beta(0.5, 0.5)						1147.4
U3 - Beta(0.7, 0.3)						1157.6
U3 - Beta(1, 1)						1145.6
Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	9.7	39.7	17.6	8.9	2	22.1
U2 - Beta(0.5, 0.5)	12.2	41.6	16.1	7.9	1.7	20.5
U2 - Beta(0.7, 0.3)	13	40.3	16.6	7.7	2.1	20.3
U2 - Beta(1, 1)	14	39.5	15.7	8	2	20.8
U3 - Beta(0.3, 0.7)	12	40.7	16.3	6.5	1.6	22.9
U3 - Beta(0.5, 0.5)	16	43.1	11.6	7.6	0.8	20.9
U3 - Beta(0.7, 0.3)	16.5	42.8	11.7	6.9	1	21.1
U3 - Beta(1, 1)	17.2	42.4	11.8	6.4	1	21.2
Mean administrations						
U2 - Beta(0.3, 0.7)	7.7	11.2	5.4	2.6	0.6	-
U2 - Beta(0.5, 0.5)	7.5	9.8	5.6	3.6	1.2	-
U2 - Beta(0.7, 0.3)	7.4	9.5	5.6	3.7	1.4	-
U2 - Beta(1, 1)	7.4	10	5.6	3.5	1.2	-
U3 - Beta(0.3, 0.7)	8.1	11.6	5.1	2.2	0.5	-
U3 - Beta(0.5, 0.5)	7.8	10	5.4	3.4	1.1	-
U3 - Beta(0.7, 0.3)	7.8	9.8	5.4	3.5	1.1	-
U3 - Beta(1, 1)	7.8	10.1	5.3	3.4	1	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.8	5.2	1.8	0.9	0.2	-
U2 - Beta(0.5, 0.5)	1.6	4.8	1.7	1.3	0.4	-
U2 - Beta(0.7, 0.3)	1.6	4.7	1.7	1.4	0.4	-
U2 - Beta(1, 1)	1.6	4.8	1.7	1.3	0.4	-
U3 - Beta(0.3, 0.7)	2	5.3	1.8	0.8	0.2	-
U3 - Beta(0.5, 0.5)	1.8	4.8	1.7	1.2	0.3	-
U3 - Beta(0.7, 0.3)	1.8	4.8	1.6	1.3	0.3	-
U3 - Beta(1, 1)	1.8	4.9	1.6	1.2	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						963.3
U2 - Beta(0.5, 0.5)						976.9
U2 - Beta(0.7, 0.3)						979.7
U2 - Beta(1, 1)						979.4
U3 - Beta(0.3, 0.7)						957
U3 - Beta(0.5, 0.5)						974.4
U3 - Beta(0.7, 0.3)						975.8
U3 - Beta(1, 1)						973.2

Phase I/II Operating Characteristics

Beta-binomial efficacy prior comparison - shared alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Beta(0.3, 0.7)	11.5	48.2	15.9	5	0.8	18.6
U2 - Beta(0.5, 0.5)	12.2	48.8	16.7	4.1	0.8	17.4
U2 - Beta(0.7, 0.3)	12.7	48.8	16.5	4.6	0.8	16.6
U2 - Beta(1, 1)	14.4	46.9	16.1	4.6	0.7	17.3
U3 - Beta(0.3, 0.7)	14.2	50.2	13.8	2.9	0.4	18.5
U3 - Beta(0.5, 0.5)	16	50.2	13.5	2.6	0.5	17.2
U3 - Beta(0.7, 0.3)	17.1	48.6	13.8	3.4	0.5	16.6
U3 - Beta(1, 1)	18.3	48	11.8	3.2	0.5	18.2
Mean administrations						
U2 - Beta(0.3, 0.7)	7.5	12.4	5.6	2.3	0.4	-
U2 - Beta(0.5, 0.5)	7.4	11.2	5.8	3	0.9	-
U2 - Beta(0.7, 0.3)	7.5	10.9	5.8	3.2	1	-
U2 - Beta(1, 1)	7.4	11.1	5.8	3.1	0.9	-
U3 - Beta(0.3, 0.7)	7.9	12.9	5	2	0.4	-
U3 - Beta(0.5, 0.5)	7.8	11.2	5.6	2.9	0.8	-
U3 - Beta(0.7, 0.3)	7.8	11.1	5.6	3	0.8	-
U3 - Beta(1, 1)	7.9	11.2	5.4	2.9	0.8	-
Mean IPDE administrations						
U2 - Beta(0.3, 0.7)	1.7	5.6	1.9	0.9	0.2	-
U2 - Beta(0.5, 0.5)	1.6	5.4	1.8	1.1	0.3	-
U2 - Beta(0.7, 0.3)	1.6	5.3	1.7	1.2	0.3	-
U2 - Beta(1, 1)	1.6	5.3	1.8	1.2	0.3	-
U3 - Beta(0.3, 0.7)	1.9	5.8	1.7	0.8	0.1	-
U3 - Beta(0.5, 0.5)	1.8	5.3	1.7	1.1	0.3	-
U3 - Beta(0.7, 0.3)	1.8	5.3	1.7	1.1	0.3	-
U3 - Beta(1, 1)	1.8	5.4	1.7	1.1	0.3	-
Mean trial duration (days)						Days
U2 - Beta(0.3, 0.7)						982.4
U2 - Beta(0.5, 0.5)						993.5
U2 - Beta(0.7, 0.3)						996
U2 - Beta(1, 1)						994.2
U3 - Beta(0.3, 0.7)						977.4
U3 - Beta(0.5, 0.5)						992.2
U3 - Beta(0.7, 0.3)						991.2
U3 - Beta(1, 1)						989.6

Fixed priors: toxicity additive alpha Beta(0.15, 0.85); efficacy additive alpha Beta(0.15, 0.85); efficacy threshold = 0.20.

All four beta-binomial efficacy priors use futility cutoff = 0.85; Beta(0.5, 0.5) rows are shaded.